

ONLINE ORDERING AND RESERVATION SYSTEM FOR COMPUTER SHOP

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University of Colombo School of Computing

Declaration

I certify that this dissertation does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any University/Institute, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person or myself except where due reference is made in the text.



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Abstract

The Online Ordering and Reservation System for Computer Store is a comprehensive web-based platform designed to meet the evolving needs of the computer retail industry. The project aims to address the limitations of existing systems by providing an efficient and user-friendly solution for customers and computer store owners. The system offers a range of features including product browsing, searching, sorting, and filtering, as well as a seamless checkout process with integrated payment processing. Additionally, customers can utilize the innovative build-your-own-PC feature to customize and assemble their desired computer systems. The system also incorporates effective order management and administration tools, including sales reporting and analytics. Furthermore, customers can track the status of their PC repairs in real-time and receive SMS notifications upon completion. By leveraging a variety of web technologies, the project delivers a robust and scalable system accessible on multiple devices. Overall, the Online Ordering and Reservation System for Computer Store aims to revolutionize the computer retail sector, providing an enhanced online shopping experience for customers and streamlining operations for store owners.

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List of Acronyms

PC	- Personal Computer
HTML	- Hypertext Markup Language
CSS	- Cascade Style Sheet
UI	- User Interface
API	- Application Programming Interface

Chapter 01 - Introduction

1.1. Background & Motivation for the Project

The computer retail industry has witnessed significant growth and transformation in recent years, driven by advancements in technology and changing consumer preferences [1]. With the rise of e-commerce, online purchasing has become increasingly popular, offering customers convenience and a wide range of options. However, the computer retail sector still faces challenges in providing effective online ordering and reservation systems. The absence of a user-friendly and comprehensive solution has hindered the industry's ability to meet the increasing demand for convenient and streamlined online shopping experiences. This has created a crucial gap that needs to be addressed to enhance customer satisfaction and improve operational efficiency for computer store owners.

Traditionally, customers have relied on physical visits to computer stores to purchase or reserve products. However, this approach often leads to inconvenience, especially for those with busy schedules or limited access to physical stores. Furthermore, the lack of a transparent and efficient system to track repair requests has caused frustration and uncertainty among customers. Recognizing these challenges, there is a pressing need to develop an Online Ordering and Reservation System specifically tailored to the requirements of the computer retail industry. Such a system would enable customers to browse and select products, place orders, make payments securely, and reserve items for repair seamlessly, all from the comfort of their own homes or offices.

In addition to meeting the demands and expectations of customers, computer store owners also face their own set of challenges. Manual order management, inventory tracking, and reservation administration processes can be time-consuming and prone to errors. Without a centralized system, store owners struggle to efficiently manage their operations, leading to delays, miscommunication, and loss of potential sales. By implementing an integrated Online Ordering and Reservation System, computer store owners can streamline their processes, gain valuable insights into customer preferences and behavior, and improve overall business performance. This system would not only enhance customer satisfaction

but also provide a competitive advantage for computer stores in a rapidly evolving and highly competitive market.

The rapid advancement of technology has significantly influenced the way consumers purchase products, shifting the traditional retail landscape to online platforms. The computer retail sector is no exception, with an increasing number of customers opting for online shopping due to its convenience and accessibility. However, the existing online systems in the computer retail industry often lack essential features and user-friendly interfaces, leading to a suboptimal user experience. Recognizing this gap, the Online Ordering and Reservation System for Computer Store project aims to address these shortcomings and provide an innovative and efficient solution for customers and computer store owners.

The computer retail industry is highly competitive, with numerous retailers vying for market share. In this dynamic environment, an effective online ordering system becomes imperative for computer store owners to attract customers and streamline their operations [2]. Existing systems often lack essential functionalities such as product customization, comprehensive search and filtering options, and efficient order management tools. These limitations pose challenges for both customers and administrators, hindering the overall efficiency of the online purchasing process. The proposed project seeks to bridge this gap by developing a robust and user-friendly system that incorporates these essential features, providing customers with an enhanced shopping experience and enabling store owners to manage their operations more effectively.

Moreover, the proliferation of mobile devices and the increasing popularity of e-commerce have led to a surge in online shopping through smartphones and tablets. It is crucial for the Online Ordering and Reservation System for Computer Store to be accessible and optimized for various devices, ensuring a seamless user experience regardless of the platform used. By designing a responsive and mobile-friendly web-based platform, the project aims to cater to the growing demand for mobile shopping and provide customers with the flexibility to browse and purchase products on the go.

Additionally, the proposed system recognizes the importance of customer communication and transparency in the computer retail sector. Many customers encounter difficulties in

tracking the status of their repair orders, leading to frustration and dissatisfaction. The project seeks to alleviate this issue by incorporating a dedicated feature that allows customers to monitor the progress of their PC repairs in real-time. Furthermore, the system will utilize SMS notifications to inform customers promptly when their repairs are completed, ensuring a seamless and efficient repair management process.

The motivation behind this project stems from the increasing demand for convenient and efficient online purchasing solutions in the computer retail industry. With the rapid advancement of technology and the widespread adoption of e-commerce platforms, customers now expect seamless and user-friendly online shopping experiences. However, the computer retail sector has lagged behind in providing a dedicated online ordering and reservation system that caters specifically to the unique requirements of this industry. This lack of an effective solution has created a gap that needs to be filled, prompting the need for the development of an innovative Online Ordering and Reservation System for computer stores.

The motivation for this project also arises from the desire to enhance customer convenience and satisfaction. By implementing a robust online ordering and reservation system, customers can easily browse through a wide range of computer products, place orders, make secure payments, and reserve items for repair, all from the comfort of their own homes or offices. This eliminates the need for physical visits to the store, saving time and effort for customers. The system will also provide transparency and real-time updates, enabling customers to track the status of their repair requests and receive notifications when the repairs are completed. By offering such convenience and peace of mind, the project aims to significantly improve the overall customer experience in the computer retail industry.

Furthermore, this project is motivated by the need to enhance operational efficiency for computer store owners. Manual order management, inventory tracking, and reservation administration processes can be labor-intensive and prone to errors. By implementing an integrated online system, store owners can streamline their operations, automate processes, and improve inventory management. The system will provide a centralized platform for order and reservation management, reducing manual tasks and enabling store owners to

allocate their resources more efficiently. This will result in improved productivity, reduced costs, and enhanced business performance for computer store owners.

Another motivation for this project is the potential to gain a competitive edge in the computer retail industry. With the increasing reliance on online platforms for purchasing and reservation purposes, computer stores that offer a seamless and user-friendly online ordering experience have a greater chance of attracting and retaining customers. By developing an advanced Online Ordering and Reservation System, computer store owners can position themselves as industry leaders, providing a competitive advantage over their competitors. This project aims to empower computer stores to adapt to the evolving retail landscape, meet customer expectations, and stay ahead in an increasingly digital world.

These paragraphs highlight the motivations behind your project, including the need to meet customer expectations, enhance operational efficiency, and gain a competitive advantage in the computer retail industry. They emphasize the importance of providing a dedicated online ordering and reservation system that caters to the unique requirements of this sector.

1.2. Objectives

- ✚ OBJ 01 - Develop a user-friendly and efficient online ordering platform.
- ✚ OBJ 02 - Implement a comprehensive PC customization feature.
- ✚ OBJ 03 - Enable efficient order management and administration.
- ✚ OBJ 04 - Enhance customer communication and repair tracking.

1.3. Scope

The scope of this project encompasses the design and development of an Online Ordering and Reservation System for a computer store. The system aims to provide customers with a seamless online shopping experience, including functionalities such as browsing computer products, placing orders, making secure payments, and reserving items for repair. The system will also facilitate efficient reservation administration for store owners, enabling them to effectively manage and track reservations, ensuring accurate inventory management.

1.3.1. In Scope

Table 1.1 : List of In Scope

SCP 1	User registration and login functionality
SCP 2	Product browsing, search, sorting, and filtering options
SCP 3	Add-to-cart and checkout process
SCP 4	Payment gateway development and integration
SCP 5	Build-your-own-PC feature with component selection and compatibility checks
SCP 6	Order management and administration tools
SCP 7	Sales reporting and analytics
SCP 8	PC repair status tracking
SCP 9	SMS notifications for order confirmations and repair completion
SCP 10	Responsive and mobile-friendly design
SCP 11	Database management for storing product information, customer data, and order details
SCP 12	Integration of web technologies such as HTML, CSS, JavaScript, PHP, and MySQL

1.3.2. Out of Scope

Table 1.2 : List of Out of Scope

OSP 1	Physical delivery and logistics management of the ordered products.
OSP 2	Hardware repair services beyond the scope of the reservation administration module.
OSP 3	Integration with external systems or third-party applications.
OSP 4	Customer support beyond the scope of the online ordering and reservation system.
OSP 4	In-depth inventory management and tracking beyond the needs of reservation administration.

It is essential to establish clear boundaries and define the in-scope and out-of-scope aspects of the project to ensure a focused and successful implementation of the Online Ordering and Reservation System for a computer store.

1.4. Structure of the Dissertation

The structure of this thesis follows a systematic and logical progression, with each chapter contributing to the overall understanding and implementation of the Online Ordering and Reservation System for a computer store. The upcoming chapters will delve into the analysis, design, implementation, evaluation, and conclusion of the project. The analysis chapter will identify the requirements and challenges faced by the computer store, laying the foundation for the subsequent design phase. The design chapter will outline the technical specifications and architectural design of the system, guiding its implementation. The implementation chapter will provide insights into the practical development process, while the evaluation chapter will assess the system's effectiveness and efficiency. Finally, the conclusion chapter will summarize the key findings and contributions, emphasizing the project's significance and potential future research areas. Together, these chapters form a comprehensive framework for understanding and implementing an advanced Online Ordering and Reservation System tailored to the unique needs of the computer retail industry.

Chapter 2: Analysis

In this chapter, a comprehensive analysis of the requirements and specifications of the Online Ordering and Reservation System for the computer store will be presented. This analysis will involve gathering information from stakeholders, including customers and store owners, to identify their needs and expectations. The chapter will delve into the functional and non-functional requirements of the system, considering aspects such as user interface design, security, performance, and scalability. Additionally, the existing processes and challenges faced by the computer store in managing orders and reservations will be analyzed to provide a solid foundation for the system's design and implementation.

Chapter 3: Design

The design chapter will outline the architectural design and technical specifications of the Online Ordering and Reservation System. It will present the overall system architecture, including the client-side and server-side components. The user interface design, database

schema, and system flow will be described in detail. The chapter will also discuss the selection of appropriate technologies and frameworks to implement the system, ensuring its scalability, security, and usability. Furthermore, the chapter will cover the design considerations for integrating the reservation administration module, payment processing, and notification system seamlessly into the system's architecture.

Chapter 4: Implementation

This chapter will focus on the practical implementation of the Online Ordering and Reservation System. It will describe the development process, including the coding, testing, and deployment stages. The implementation details of each system module, such as product browsing, ordering, payment processing, reservation administration, and notification system, will be presented. Additionally, any challenges or obstacles encountered during the implementation phase and their corresponding solutions will be discussed. Code snippets and screenshots may be provided to illustrate key functionalities and system components.

Chapter 5: Evaluation

The evaluation chapter will assess the effectiveness and efficiency of the implemented Online Ordering and Reservation System. This evaluation will be conducted through various means, including user feedback, performance testing, and system metrics analysis. The chapter will present the results of user satisfaction surveys, highlighting any improvements and enhancements suggested by the users. Performance metrics, such as system response time and throughput, will be measured to ensure the system's scalability and reliability. The evaluation will provide valuable insights into the system's strengths, weaknesses, and areas for future improvement.

Chapter 6: Conclusion

The conclusion chapter will summarize the key findings and outcomes of the thesis. It will revisit the objectives of the project and evaluate whether they have been successfully achieved. The chapter will highlight the contributions of the Online Ordering and Reservation System in addressing the challenges faced by the computer retail industry and improving customer satisfaction and operational efficiency. Additionally, it will discuss the

significance of the project, its implications for the field, and potential avenues for future research and development.

Appendices

The appendices will include supplementary information that supports and complements the main content of the thesis. This may include the detailed system requirements, user interface design prototypes, database schema, sample code snippets, and any additional technical documentation. The appendices provide a reference for readers who seek more in-depth information about specific aspects of the Online Ordering and Reservation System.

Overall, these chapters will provide a comprehensive understanding of the project, from the initial analysis and design stages to the practical implementation, evaluation, and final conclusions. The thesis will present a holistic view of the Online Ordering and Reservation System and its significance in improving the computer retail industry's operations and enhancing customer experiences.

Chapter 02 - Analysis

The Analysis chapter serves as a critical component of this thesis, aiming to thoroughly examine the requirements and challenges associated with the development of an Online Ordering and Reservation System for a computer store. By delving into the needs and expectations of both customers and store owners, this chapter lays the groundwork for the subsequent design and implementation phases. Through stakeholder interviews, surveys, and extensive research, this analysis will identify the key functional and non-functional requirements of the system, ensuring that it addresses the specific pain points and demands of the computer retail industry. Additionally, a comprehensive assessment of the existing order and reservation management processes will be conducted, uncovering inefficiencies and areas for improvement. By conducting this in-depth analysis, we can establish a solid foundation for the subsequent stages of the project, ultimately leading to the development of an innovative and effective solution.

2.1. Existing Systems

Existing systems in the computer retail industry encompass a range of online platforms utilized by customers and computer store owners. However, many of these systems suffer from limitations that hinder the overall user experience and operational efficiency. Some existing systems lack essential features such as online checkout functionality, product customization options, and comprehensive search and filtering capabilities, as shown in the use case diagram in figure 2.1. Additionally, user interfaces are often outdated and lack visual appeal, leading to a less engaging and user-friendly experience. Furthermore, customer communication and repair tracking features are often absent, resulting in a lack of transparency and customer satisfaction. Recognizing these limitations, the Online Ordering and Reservation System for Computer Store project aims to address these gaps by providing a robust and innovative solution that enhances the online shopping experience for customers and streamlines operations for store owners.

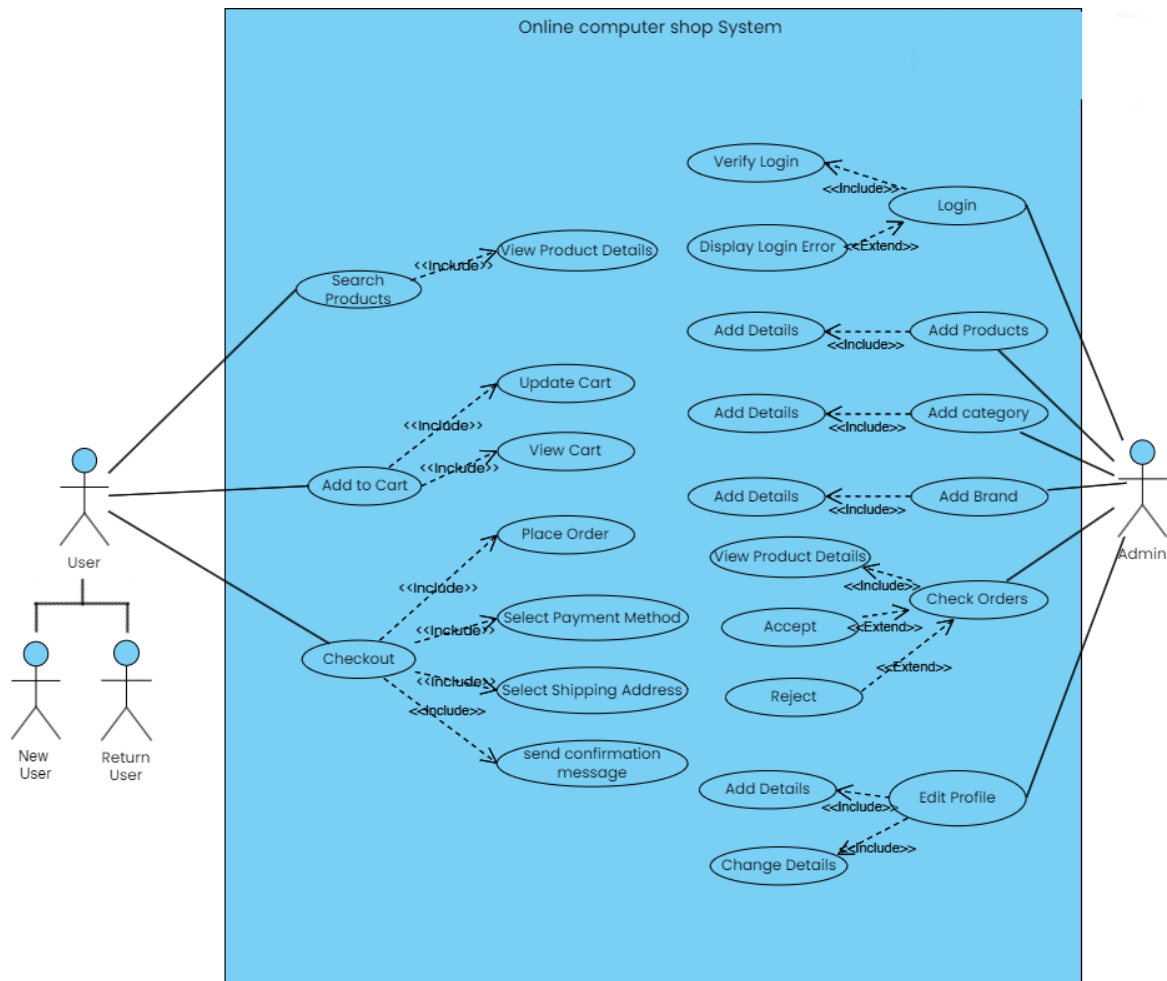


Figure 2.1 : Use-Case Diagram of an Existing Computer Shop System

2.2. Drawbacks of the Existing Systems

- Lack of online checkout functionality, requiring customers to complete purchases through alternative means or in-person.
- Limited or outdated user interfaces that are not visually appealing or user-friendly.
- Absence of product customization options, limiting customers' ability to personalize their computer systems.
- Inadequate search and filtering capabilities, making it difficult for customers to find specific products based on their preferences.
- Insufficient communication features, leading to challenges in customer support and inquiries.
- Lack of transparency in repair tracking, resulting in customer frustration and dissatisfaction.

- Incomplete or inaccurate product information, making it challenging for customers to make informed purchasing decisions.
- Inefficient order management processes, leading to delays or errors in processing customer orders.
- Limited reporting and analytics functionalities, hindering store owners' ability to gather insights and make informed business decisions.
- Incompatibility with mobile devices, making it inconvenient for customers to browse and shop on smartphones or tablets.

These drawbacks highlight the areas where existing systems in the computer retail industry often fall short, emphasizing the need for a more comprehensive and improved solution.

2.3. Review of Similar Systems

In the review of similar systems, this thesis will analyze and evaluate three existing web-based online platforms in the computer retail industry. These platforms have been selected to provide a comprehensive understanding of the current landscape and identify the strengths and weaknesses of each system. By examining and comparing these systems, this thesis aims to identify opportunities for improvement and inform the development of the Online Ordering and Reservation System for Computer Store.

2.3.1. Gamestreet.lk

Gamestreet.lk is an existing online platform in the computer retail industry [3]. However, it lacks a checkout feature, making it inconvenient for customers to complete purchases directly on the website. The user interface (UI) is considered poor, which may result in a less engaging and user-friendly experience. Figure 2.2 shows a screenshot of the web page. Additionally, there is no provision to track the status of repairs or notify customers via SMS when repairs are completed. The absence of checkout confirmation and build-your-own-PC features further limits the platform's functionality. User accounts, filtering options, and generating reports are also not available.

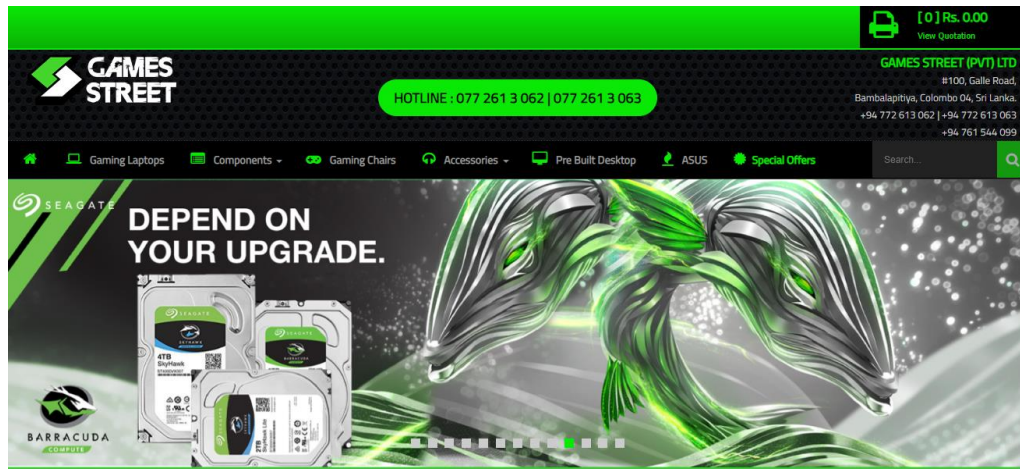


Figure 2.2 : Screenshot of Gamestreet.lk

2.3.2. Nexxcom.lk

Nexxcom.lk is another existing online platform in the computer retail industry [4]. It offers a checkout option, allowing customers to complete purchases online. However, the UI is considered subpar, which may impact the overall user experience. Similar to Gamestreet.lk, Nexxcom.lk lacks a repair status tracking feature and does not send SMS notifications when repairs are completed or for checkout confirmation. While user accounts are available, the platform lacks filtering options and generating reports functionalities. Figure 2.3 shows a screenshot of this web page.

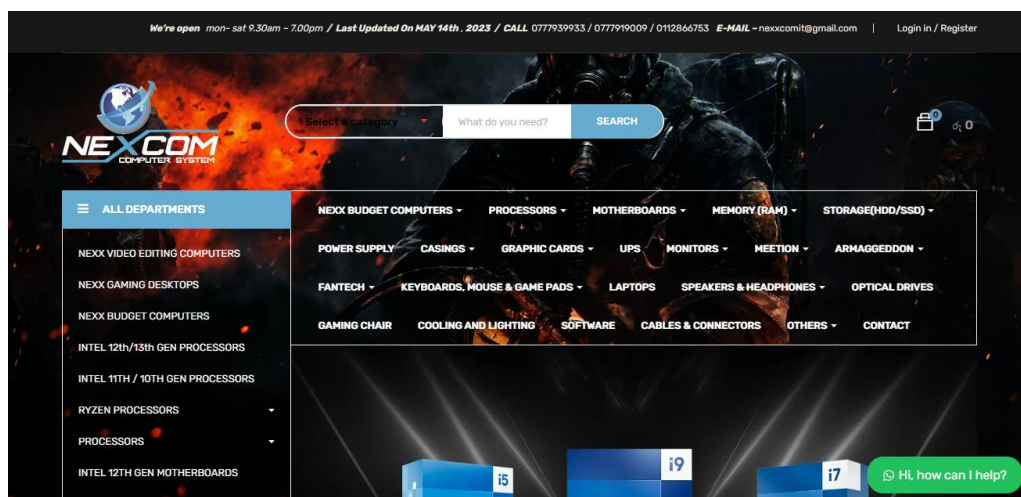


Figure 2.3 : Screenshot of Nexxcom.lk

2.3.3. Nanotek.lk

Nanotek.lk is an existing online platform that offers various features in the computer retail industry [5]. However, it does not provide a checkout option, requiring customers to explore alternative methods to complete their purchases. On a positive note, the UI is highly regarded, offering an attractive and user-friendly experience. Similar to the previous systems, Nanotek.lk lacks repair status tracking and SMS notifications for repair completion and checkout confirmation. However, it does offer a build-your-own-PC feature, filtering options, and generating reports functionalities. User accounts are not available on this platform.

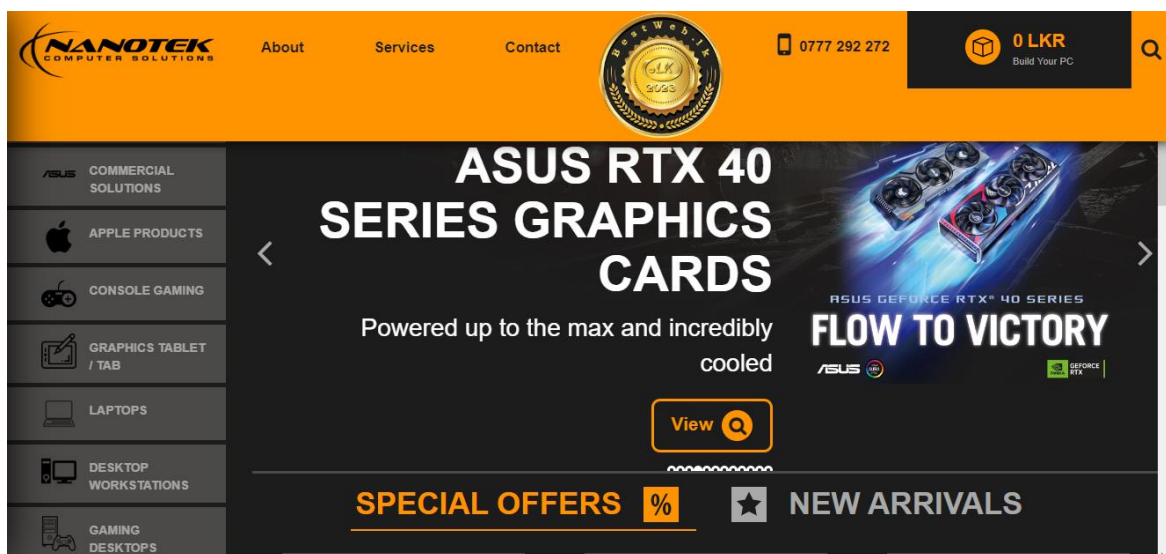


Figure 2.4 : Screenshot of Nanotek.lk

2.3.4. Comparison Between Similar Systems

Table 2.1 : Comparison of Existing Computer Shop Platforms

Component / Feature	Gamestreet.lk	Nexxcom.lk	Nanotek.lk	Proposed System
Online Checkout	✗	✓	✗	✓
High Quality & User-Friendly UI	✗	✗	✓	✓
Repair Status Tracking	✗	✗	✗	✓
Product Catalog	✓	✓	✓	✓
SMS Notifications for Repair Completion	✗	✗	✗	✓
Checkout Confirmation SMS	✗	✗	✗	✓
Build Your PC Feature	✗	✗	✓	✓
User Accounts	✗	✓	✗	✓
Filtering Options	✗	✗	✓	✓
Generating Sales Reports	✗	✗	✓	✓

2.4. Requirement Analysis Techniques

The requirement gathering techniques play a crucial role in the success of any software development project. This section focuses on exploring various techniques employed to gather the necessary requirements for the Online Ordering and Reservation System for Computer Store. Effective requirement gathering ensures that the system is designed and developed to meet the needs and expectations of its users. By employing a combination of techniques, including interviews, surveys, and observation, this project aims to capture and analyze the requirements from different perspectives. The goal is to gather comprehensive

and accurate information that will serve as the foundation for the subsequent phases of analysis, design, and implementation. This section delves into the techniques utilized to gather, validate, and prioritize the requirements, highlighting their importance in shaping the development process and ensuring the final system aligns with the expectations of both customers and computer store owners.

2.4.1. Interviews

Conducting one-on-one interviews with key stakeholders, including computer store owners, customers, and system administrators, to gather their insights, expectations, and specific requirements for the system was done for this project.

2.4.2. Surveys

The project involved conducting two surveys to gather valuable insights from customers and computer shop owners. The Customer Survey shown in figure 2.5 focused on understanding online shopping preferences, experiences, and expectations related to computer products. It aimed to gather feedback on features such as user accounts, checkout process, repair status tracking, and SMS notifications. On the other hand, the Computer Shop Owner Survey shown in figure 2.6 aimed to gather information about the importance of an online ordering system, challenges faced in managing online orders and reservations, and preferences for features like repair status tracking and sales reports. Both surveys aimed to collect data that would inform the development of an enhanced Online Ordering and Reservation System for a Computer Store.

Questions Responses 114 Settings

Customer Survey: Online Shopping Preferences and Experiences for Computer Products

Thank you for taking the time to participate in our survey! Your feedback is valuable in helping us develop an enhanced Online Ordering and Reservation System for Computer Store. This survey aims to gather insights into your online shopping preferences, expectations, and experiences related to computer products. Your responses will remain anonymous and confidential. Please answer the following questions to the best of your knowledge and feel free to provide any additional comments or suggestions at the end. Your input will greatly contribute to creating a user-friendly and efficient platform for your future computer-related purchases. Thank you for your participation!

How frequently do you purchase computer products online? *

☐ a) Rarely

☐ b) Occasionally

☐ c) Frequently

Figure 2.5 : Customer Survey

Questions Responses 54 Settings

Computer Shop Owner Survey

Thank you for participating in our survey! Your insights and feedback are vital in helping us understand the needs and challenges faced by computer shop owners like you. This survey aims to gather valuable information about your experience with online orders and reservations, as well as your preferences and requirements for an effective Online Ordering and Reservation System for Computer Store. Your responses will be kept confidential and will only be used for research purposes. Please take a few moments to answer the following questions, and feel free to provide any additional comments or suggestions at the end. Your input will greatly contribute to the development of a tailored system that meets the specific needs of computer shop owners. Thank you for your valuable contribution!

How important is it for your computer shop to have an online ordering system? *

☐ a) Not Important

☐ b) Somewhat Important

☐ c) Moderately Important

Figure 2.6 : Shop Owner Survey

2.4.3. Observation

The existing processes and interactions in computer stores were observed and analyzed to gain a better understanding of the workflow, pain points, and areas where improvements and efficiencies can be brought by the system.

2.4.4. Documentation Analysis

The relevant documentation has been reviewed, including existing system documentation, industry standards, and best practices, in order to identify essential requirements and gather insights for system enhancements.

2.5. Requirement Analysis

The requirement analysis section focuses on thoroughly examining and understanding the needs, expectations, and specifications for the development of the Online Ordering and Reservation System for Computer Store. This section delves into the gathered data from surveys, interviews, and documentation review, aiming to extract and analyze the essential requirements and features desired by both customers and computer shop owners. By conducting a comprehensive requirement analysis, we can ensure that the system's design and functionality align with the identified needs and provide an enhanced online shopping experience.

2.5.1. Functional Requirements

Table 2.2 presents the functional requirements of the Online Ordering and Reservation System for Computer Store.

Table 2.2 : Functional Requirements

	Requirement	Description
FR 01	User Registration and Login	Customers can create user accounts to access personalized features. Computer shop owners can have separate accounts with administrative privileges.
FR 02	Product Browsing and Search	Users can browse and search for computer products based on categories, brands, and specifications. Advanced search options allow users to filter products based on specific criteria.
FR 03	Product Details	Users can view detailed information about each product, including specifications, prices, and availability.
FR 04	Add to Cart and Checkout	Users can add selected products to their shopping cart. A secure checkout process allows users to complete their purchases, providing multiple payment options.
FR 05	Build Your Own PC	Users can customize and build their own PC configurations by selecting compatible components.
FR 06	Order Management	Computer shop owners can view and manage incoming orders, including order confirmation and tracking.
FR 07	Repair Status Tracking	Users can check the status of their product repairs online, providing real-time updates.
FR 08	SMS Notifications	Users receive SMS notifications for order confirmations, repair status updates, and completed repairs.
FR 09	Sales Reports and Analytics	Computer shop owners can generate comprehensive sales reports and analytics to track performance and make informed business decisions.
FR 10	Admin Dashboard	Computer shop owners have access to an administrative dashboard to manage product inventory, user accounts, and system settings.

2.5.2. Non-Functional Requirements

Table 2.3 presents the non-functional requirements of the Online Ordering and Reservation System for Computer Store.

Table 2.3 : Non-Functional Requirements

	Requirement	Description
NFR 01	Usability	The system should have an intuitive and user-friendly interface to ensure ease of navigation and a positive user experience.
NFR 02	Performance	The system should have fast response times to handle multiple user interactions simultaneously. The system should be able to handle a large number of users and transactions without compromising performance.
NFR 03	Security	The system should implement robust security measures, including encryption of sensitive data and secure payment processing. User authentication and authorization mechanisms should be in place to protect user accounts and prevent unauthorized access.
NFR 04	Reliability	The system should be highly reliable, with minimal downtime and interruptions in service. It should have backup and recovery mechanisms in place to ensure data integrity and continuity of operations.
NFR 05	Scalability	The system should be designed to accommodate future growth and increased user demand without significant performance degradation. It should be scalable to handle a growing product inventory and expanding customer base.
NFR 06	Integration	The system should be able to integrate with external systems or APIs, such as payment gateways and shipping services, to facilitate seamless transactions and order fulfillment.
NFR 07	Compliance	The system should adhere to relevant legal and industry regulations, such as data protection and privacy laws. It should comply with standards and best practices for e-commerce platforms and online payment processing.
NFR 08	Maintenance and Support	The system should be easy to maintain, with clear documentation and provisions for future updates and enhancements. Adequate technical support should be available to address user inquiries and resolve any system-related issues.

2.5.3. Objective Requirement mapping

Table 2.4 presents the mapping of project objectives to the corresponding functional requirements of the Online Ordering and Reservation System for Computer Store. This mapping highlights the direct relationship between each objective and the specific requirements that are essential for achieving it. By aligning the objectives with the associated requirements, this chart provides a clear overview of how the system's functionalities and qualities contribute to the fulfillment of the project's objectives. It serves as a valuable reference for project stakeholders and developer to ensure that the system design and implementation are focused on meeting the defined objectives effectively.

Table 2.4 : Object Requirement Mapping Chart

Objective	Functional Requirement
OBJ 01	FR 01, FR 02, FR 03, FR 04
OBJ 02	FR 05
OBJ 03	FR 06, FR 09, FR 10
OBJ 04	FR 07, FR 08

2.6. Justification For the Choice of the Development Life Cycle

The development life cycle is a critical decision that shapes the overall approach and methodology for a project. In the case of the Online Ordering and Reservation System for Computer Store, the modified Waterfall model shown in figure 2.7 has been chosen as the preferred development life cycle. This section provides a justification for this choice based on the specific characteristics of the project and its requirements.

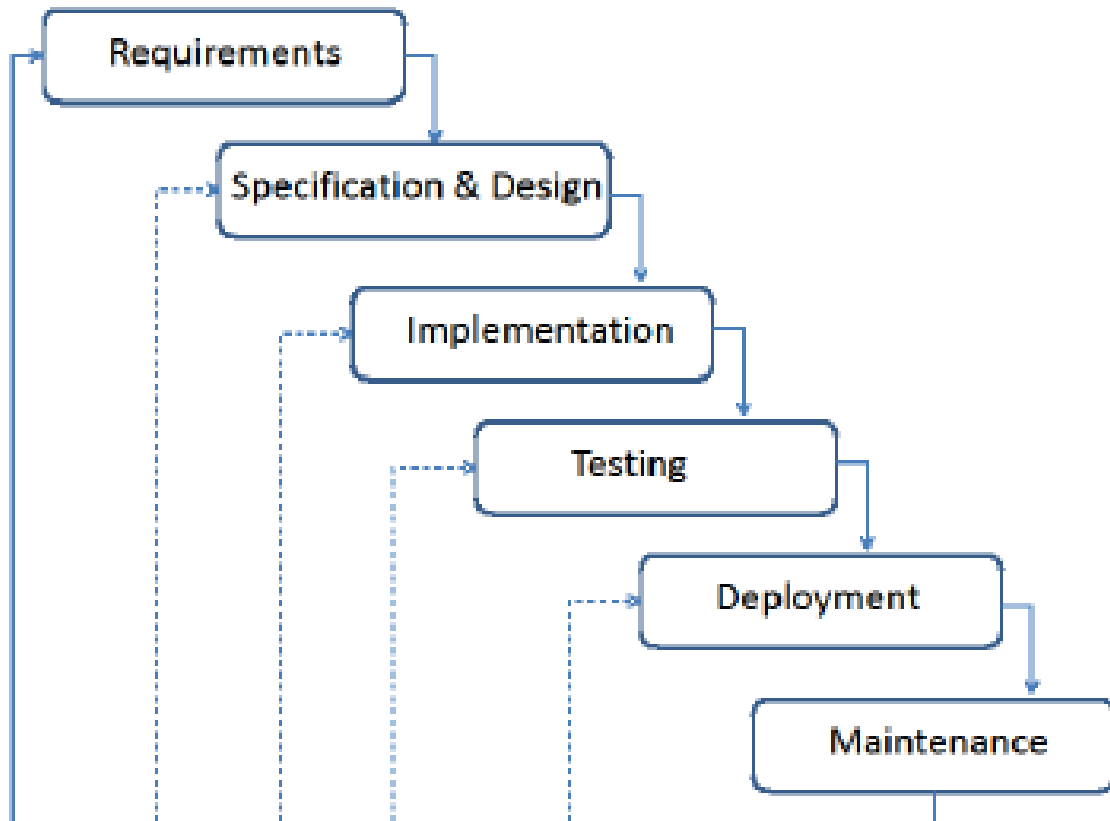


Figure 2.7 : Stages of Modified Waterfall Model

- **Stability of Requirements**

The project requirements for the Online Ordering and Reservation System are well-defined and relatively stable. There is a clear understanding of the desired functionalities and features, and minimal expectation for significant changes during the development process. The Waterfall model, with its sequential nature, aligns well with such stable requirements and allows for a structured approach to be followed.

- **Predictability and Planning**

The modified Waterfall model offers a high level of predictability and enables comprehensive planning. The project can be divided into distinct phases, including requirements analysis, design, development, testing, and deployment. Each phase has its well-defined deliverables and dependencies, making it easier to estimate timelines, allocate resources, and plan activities accordingly.

- Emphasis on Documentation

The modified Waterfall model places strong emphasis on documentation, which is essential for maintaining clarity and traceability throughout the project. Detailed documentation, such as requirements specifications, design documents, and test plans, can be created at each phase, providing a comprehensive record of the system's development. This documentation can be useful for future reference, maintenance, and enhancements.

- Clear Milestones and Progress Evaluation

The Waterfall model provides clear milestones for each phase, allowing for regular evaluation of progress. This enables the developer to monitor the project's advancement and ensure that each phase's objectives are achieved before moving on to the next. The distinct phases facilitate progress reporting and decision-making at key stages of the project.

- Limited User Involvement and External Dependencies

In this project, there may be limited direct user involvement and minimal external dependencies. The modified Waterfall model accommodates this by focusing on upfront requirements gathering and design, followed by sequential development and testing. It reduces the reliance on continuous user feedback and allows for a more independent execution of the project tasks.

While the modified Waterfall model has its strengths in providing structure, predictability, and comprehensive documentation, it is important to acknowledge that it may not be as adaptable to changes or allow for quick iterations compared to more agile approaches. However, considering the stable requirements and limited user involvement in this project, the modified Waterfall model is deemed suitable to deliver the desired outcomes efficiently.

By adopting the modified Waterfall model, the developer can ensure a systematic and well-planned approach, enabling them to focus on delivering a high-quality Online Ordering and Reservation System for Computer Store that meets the project's objectives and stakeholder expectations.

Chapter 03 - Design

The design phase of the Online Ordering and Reservation System for Computer Store is a crucial stage in the development process, where the project's requirements are translated into a concrete and detailed system design. This section presents an in-depth exploration of the design aspects, including the architectural design, database design, and user interface design. The goal is to provide a comprehensive understanding of how the system will be structured and function, ensuring that it meets the defined requirements and aligns with the project's objectives. By focusing on the design phase, the project can lay a solid foundation for the successful implementation and development of a robust and user-friendly online platform.

3.1. Alternative Solutions Considered

During the project planning phase, several alternative solutions were considered for the Online Ordering and Reservation System for Computer Store. These included manual order processing, third-party e-commerce platforms, and custom-built in-house solutions. Each option was evaluated based on factors such as scalability, customization, control, and cost-effectiveness. After careful analysis, it was determined that developing a dedicated online system tailored to the specific needs of the computer store would provide the most efficient and effective solution.

- Manual Order Processing

One alternative solution that was considered was a manual order processing system. This would involve customers placing orders through phone calls or in-person visits to the computer store. The store staff would manually record the order details, process the payment, and handle the reservation or delivery arrangements. While this approach may be suitable for smaller-scale operations, it lacks the scalability and efficiency that an automated online system can provide. It would require extensive human intervention, increasing the chances of errors and delays in order processing. Additionally, manual systems are more susceptible to inconsistencies and lack the convenience and flexibility that an online system offers.

- Third-Party E-commerce Platforms

Another alternative solution explored was utilizing third-party e-commerce platforms to handle the online ordering and reservation process. These platforms provide ready-made solutions for setting up online stores and managing transactions. While this approach offers convenience and ease of setup, it comes with limitations. Customization options may be limited, making it challenging to tailor the platform to specific business needs. There may also be restrictions on data ownership and integration with existing systems. Moreover, relying on third-party platforms means less control over the user experience and branding, which can be crucial for establishing a unique identity in the competitive computer store market.

- Custom-Built In-House Solution:

An alternative solution considered was the development of a custom-built in-house system. This approach would involve building the online ordering and reservation system entirely from scratch, tailored to the specific requirements of the computer store. While this offers maximum flexibility and control over the system's functionality and design, it requires significant time, effort, and expertise. Developing a robust and feature-rich system internally would involve a considerable investment in terms of development resources, infrastructure, and ongoing maintenance. Additionally, it may require additional training for staff members to handle the system effectively. Considering the project's timeline and available resources, a custom-built in-house solution was deemed impractical and less cost-effective compared to other alternatives.

After careful evaluation of these alternative solutions, it was determined that developing a dedicated online ordering and reservation system with customized functionalities and a user-friendly interface would best address the unique needs of the computer store. This solution allows for greater control, scalability, and flexibility while providing a tailored experience for both customers and store owners.

3.1.1. Pros and Cons of each approach

Table 3.1 : Pros and Cons of Each Approach

Approach	Pros	Cons
Manual Order Processing	• Familiar and straightforward process for store staff and customers. • Minimal upfront technology investment required.	• Prone to human errors and delays in order processing. • Limited scalability and efficiency for larger volumes of orders.
Third-Party E-commerce Platforms	• Convenient and ready-made solution for setting up an online store. • Lower development and maintenance costs compared to custom-building.	• Limited customization options and branding control. • Potential restrictions on data ownership and integration with existing systems.
Custom-Built In-House Solution	• Maximum flexibility and control over system functionality and design. • Tailored solution to meet specific business requirements.	• Significant time, effort, and expertise required for development. • Higher upfront development and infrastructure costs.
Proposed Approach: Developing a Dedicated Online Ordering and Reservation System	• Greater control, scalability, and flexibility for the computer store. • Customized functionalities to address specific needs. • Enhanced user experience and branding opportunities.	• Higher upfront development costs compared to manual order processing. • Ongoing maintenance and support required for the system.

3.1.2. Justification for the Final Selection

The final selection of developing a dedicated online ordering and reservation system was justified based on several factors. Firstly, this approach offers the computer store maximum control and flexibility over the system's functionality, design, and customization options. It allows for tailored features to address specific business requirements and provide an enhanced user experience. Secondly, while there is an upfront investment in development costs, this approach ensures long-term scalability and growth potential, accommodating increased order volumes and an expanding customer base.

Additionally, by having a dedicated system, the computer store retains full ownership and control over the data, ensuring privacy and security. Lastly, the proposed solution aligns with the strategic goals of the computer store, providing a unique branding opportunity and competitive advantage in the market. Overall, the justification for the final selection is based on the ability of the dedicated online ordering and reservation system to meet the specific needs of the computer store, offer greater control, scalability, and customization, and align with long-term business objectives.

3.2. Selected Solution

The selected solution for the project is a dedicated online ordering and reservation system for the computer store. To implement this solution, a combination of tools and technologies were utilized. For the backend development, Laravel, a popular PHP framework, was chosen to provide a robust and scalable foundation. It offers a wide range of features and functionalities to handle user authentication, order management, and system administration. On the frontend side, Vue.js, along with HTML, CSS, and jQuery, was employed to create a user-friendly and attractive interface. Vue.js is a powerful JavaScript framework that enables efficient rendering and dynamic updates, enhancing the overall user experience. As for the database, MySQL was selected as the preferred choice to store and manage the system's data. MySQL is a reliable and widely used relational database management system, offering strong data integrity and performance. By leveraging these tools and technologies,

the selected solution ensures seamless integration between the frontend and backend components, providing an intuitive user interface, efficient data management, and a reliable online ordering and reservation experience for both customers and store owners.

3.2.1. Overview of the Selected Solution's Approach and Functionality

The selected solution's approach focuses on providing a comprehensive set of functionalities to meet the requirements of the computer store's online ordering and reservation system. Here is an overview of each function.

- **User Account Management:** The system includes a user account management feature that allows customers to create accounts, log in, and access personalized features. This functionality enables customers to view order history, manage personal information, and track the status of their orders and repairs.
- **Product Browsing and Filtering:** Customers can browse the computer store's product catalog using search, sorting, and filtering options. They can easily find the desired products based on categories, brands, specifications, and price ranges. This functionality enhances the user experience by providing a convenient way to explore and discover relevant products.
- **Online Ordering and Checkout:** The system enables customers to add products to their cart, review the cart contents, and proceed to checkout. During the checkout process, customers can select preferred payment methods, enter shipping details, and confirm the order. This functionality ensures a smooth and secure online purchasing experience.
- **PC Customization (Built Your PC) :** An essential feature of the system is the PC customization functionality, allowing customers to build their own customized computers. They can select desired components such as processors, RAM, storage,

graphics cards, and more. This functionality provides flexibility for customers to tailor their computers according to their specific needs and preferences.

- **Repair Tracking:** Customers can track the status of their repairs through the system. They can submit repair requests, view the progress of ongoing repairs, and receive notifications when the repairs are completed. This functionality keeps customers informed and engaged throughout the repair process, enhancing transparency and customer satisfaction.
- **Communication and Notifications:** The system incorporates features to facilitate communication between the computer store and customers. Customers can send messages to the store regarding their orders, repairs, or general inquiries. Moreover, the system sends automated notifications to customers through SMS or email, informing them about order confirmations, repair updates, and completion notifications.
- **Order and Repair Management:** The system provides comprehensive order and repair management capabilities for the computer store. Store owners can review and confirm customer orders, manage inventory, track repair requests, assign technicians, and monitor repair progress. This functionality streamlines the store's operational processes, ensuring efficient order and repair management.

By incorporating these functionalities into the selected solution, the online ordering and reservation system offers an intuitive and user-friendly platform for customers to browse, order, and track products and repairs. It also provides efficient administrative tools for the computer store to manage orders, repairs, and inventory effectively.

3.2.2. Potential Challenges and Risks

The development and implementation of any complex system come with inherent challenges and risks. In the case of our selected solution, we have identified several potential challenges and risks that could impact the successful delivery and adoption of the system. These challenges include integration complexity, scalability and performance concerns, security and data protection, as well as user acceptance and adoption. However, we have proactively addressed these challenges by implementing appropriate solutions and mitigation strategies. By taking these factors into consideration and adopting necessary measures, we aim to minimize the impact of these challenges and risks, ensuring a successful and effective implementation of the proposed solution. Following are the main challenges identified and the solutions we discovered for them.

- Integration Complexity

To overcome the challenge of integration complexity, we adopted a modular and scalable approach in the system design. By breaking down the system into smaller components, we were able to focus on integrating each module separately, ensuring a smoother integration process. Additionally, we utilized standard APIs and frameworks to facilitate seamless communication between different system modules and third-party services, reducing the complexity of integration.

- Scalability and Performance

To address the potential challenges related to scalability and performance, we implemented a robust and scalable architecture for the system. The backend, built using Laravel (PHP), was designed to handle a high volume of concurrent users and transactions efficiently. We leveraged caching mechanisms, database optimizations, and server scaling techniques to ensure optimal performance under heavy load. Regular performance testing and optimization were conducted to identify and resolve any bottlenecks or performance issues.

- Security and Data Protection

Security and data protection were given utmost priority in the selected solution. We implemented industry-standard security measures such as data encryption, secure authentication mechanisms, and role-based access control. The system architecture followed best practices for securing sensitive user information and preventing unauthorized access. Regular security audits and vulnerability assessments were performed to identify and address any potential security risks. Additionally, we implemented regular data backups and disaster recovery plans to protect against data loss or system failures.

- User Acceptance and Adoption

To ensure user acceptance and adoption of the selected solution, we conducted thorough user testing and incorporated user feedback throughout the development process. User experience (UX) design principles were applied to create an intuitive and user-friendly interface, enhancing the usability of the system. We also provided comprehensive user documentation and conducted training sessions to facilitate user onboarding and familiarization with the system. Continuous user engagement and support channels were established to address any user concerns or issues promptly.

By proactively addressing these potential challenges and risks, we have implemented appropriate solutions and strategies to mitigate their impact on the successful implementation and adoption of the selected solution.

3.3. Use of Design Techniques

In designing the proposed system, we employed several design strategies to ensure an efficient, scalable, and user-friendly solution. One of the key design strategies used was a modular approach, dividing the system into distinct and reusable modules to promote code reusability and maintainability. This modular design enables easier maintenance and future enhancements. Additionally, we utilized object-oriented design techniques, such as encapsulation, inheritance, and polymorphism, to achieve modular and extensible code structures [6].

Another design strategy employed was the use of responsive web design principles, ensuring that the system is accessible and user-friendly across different devices and screen sizes. This approach enhances the user experience and enables seamless interaction with the system, whether on desktop computers, laptops, tablets, or smartphones [7].

Furthermore, we employed database normalization techniques to ensure efficient data organization and retrieval. By following best practices in database design, such as reducing redundancy and ensuring data integrity, we achieved an optimized database structure that supports the system's data storage and retrieval requirements [8].

Overall, the design strategies used in this project focus on promoting modularity, scalability, usability, and data efficiency. These design techniques, such as modular design, responsive web design, object-oriented design, and database normalization, contribute to the development of a robust and user-friendly system that meets the project's objectives and requirements.

3.4. Architectural Design

In the architectural design phase of the project, we focused on defining the overall structure and organization of the system. The architectural design provides a high-level view of how different components and modules of the system interact and collaborate to achieve the desired functionality.

For this project, we adopted a layered architecture approach shown in figure 3.1, which separates the system into logical layers, each with specific responsibilities.

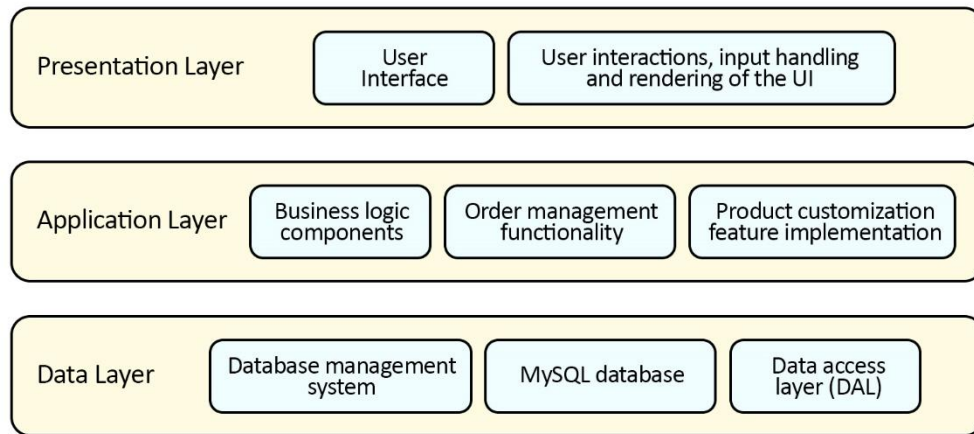


Figure 3.1 : layered architecture approach

The layered architecture promotes modularity, reusability, and maintainability by enforcing a clear separation of concerns. The typical layers include the presentation layer (frontend), the application layer (backend), and the data layer (database). This design allows for easier modification or replacement of individual layers without affecting the entire system. Within each layer, we employed appropriate design patterns and principles to guide the implementation. For example, in the presentation layer, we utilized the Model-View-Controller (MVC) pattern to separate the user interface, data, and application logic. This separation ensures a clear division of responsibilities and facilitates easier maintenance and testing. Furthermore, we considered scalability and performance in the architectural design. By distributing the system across multiple servers or implementing caching mechanisms, we aimed to ensure that the system can handle increased user load and provide optimal performance.

The Online Ordering and Reservation System for Computer Store follows the Model-View-Controller (MVC) architectural pattern to ensure a well-structured and maintainable design [9]. The MVC pattern separates the application into three distinct components: the Model, the View, and the Controller as shown figure 3.2.

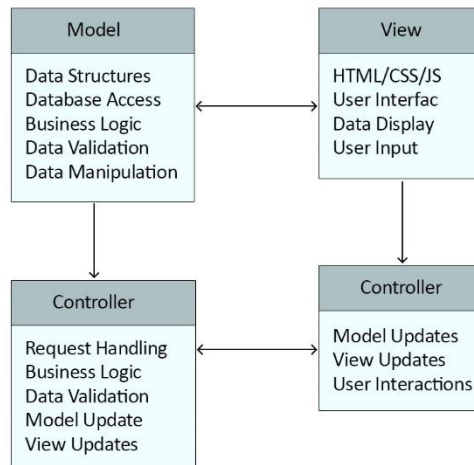


Figure 3.2 : Model-View-Controller (MVC) Architectural Pattern

The Model represents the data and business logic of the system. It encapsulates the data structures, database access, and data manipulation operations. It is responsible for handling data storage, retrieval, and validation. In the context of the project, the Model component handles tasks such as managing product information, order details, user accounts, and performing necessary database operations.

The View component focuses on the user interface and presentation logic. It includes HTML, CSS, and JavaScript to render the web pages that users interact with. The View displays the relevant information retrieved from the Model and provides a visual representation of the system's data and functionality. In the case of the project, the View component is responsible for displaying product catalogs, order forms, user account information, and other relevant pages.

The Controller acts as an intermediary between the Model and the View. It receives user input and triggers the appropriate actions in the Model and updates the View accordingly. The Controller handles request handling, business logic execution, data validation, and coordination between the Model and the View. In the project, the Controller component manages tasks such as processing user orders, updating the database, validating user input, and coordinating the communication between the Model and the View.

By adopting the MVC architecture, the Online Ordering and Reservation System for Computer Store achieves a clear separation of concerns, making the system more modular, scalable, and maintainable. The Model handles data-related operations, the View takes care

of the presentation layer, and the Controller manages the interactions and logic flow. This division of responsibilities facilitates code organization, enhances code reusability, and allows for easier maintenance and future enhancements of the system.

Overall, the architectural design of the project provides a solid foundation for the system, enabling flexibility, modularity, and scalability. It ensures that the system's components are organized and structured in a way that facilitates efficient development, maintenance, and future enhancements.

3.4.1. Use Case Diagram

The use case diagram (Figure 3.3) represents the various interactions and relationships between actors and use cases in the project. The diagram provides a visual representation of the functionalities and features that the system offers to its users. The primary actors are depicted, including the customer and the admin, who interact with the system to perform specific tasks. The use cases, such as "Logging into account," "Finding product," "Adding product to cart," and others, depict the specific actions that the actors can perform within the system. The use case diagram gives an overview of the system's functionality and serves as a foundation for further development and design.

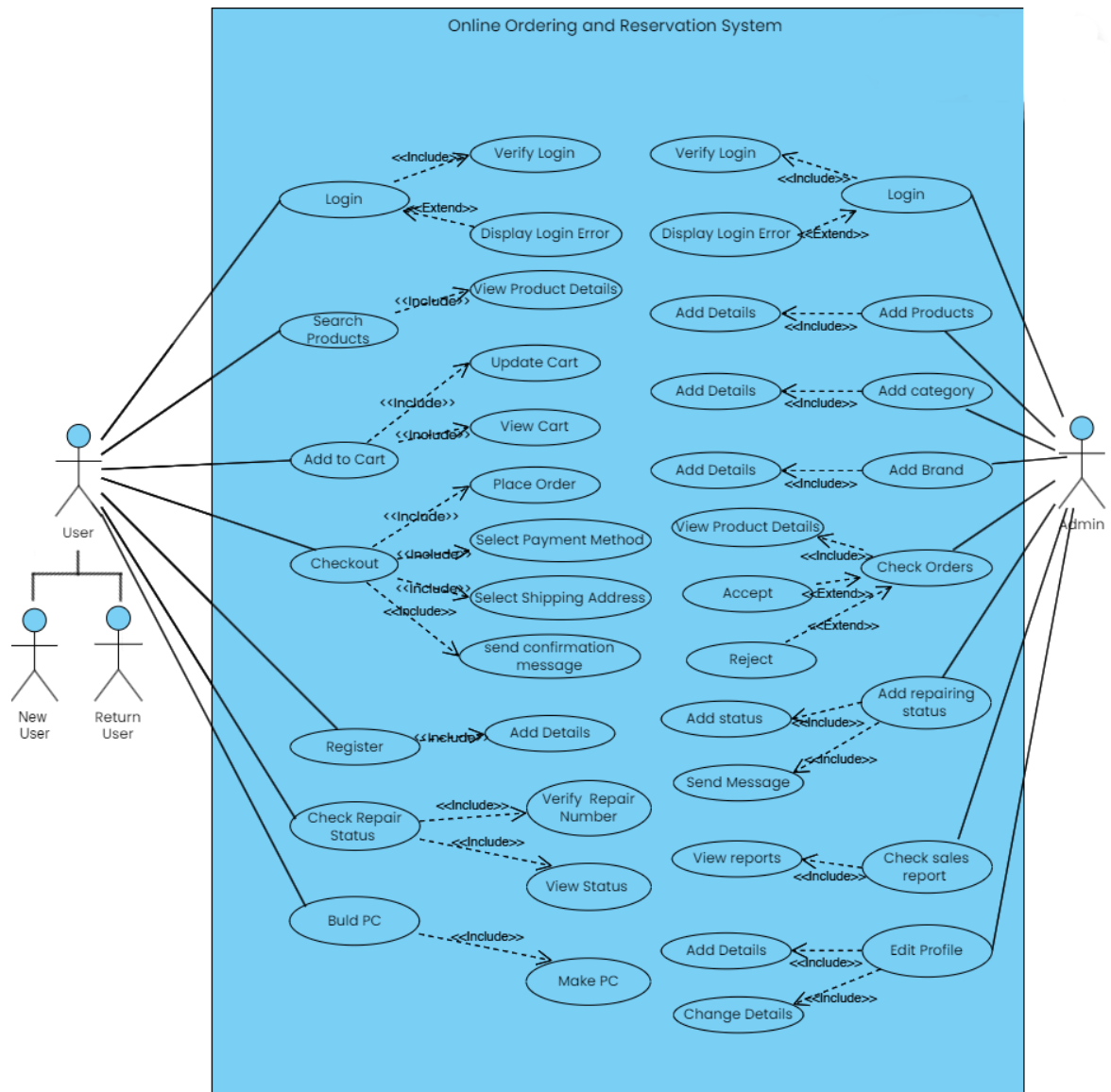


Figure 3.3 : Use-Case Diagram

3.4.2. Class Diagram

The class diagram (Figure 3.4) presents a static view of the system's structure by showcasing the classes, their attributes, and the relationships among them. It provides a comprehensive overview of the object-oriented design of the project.

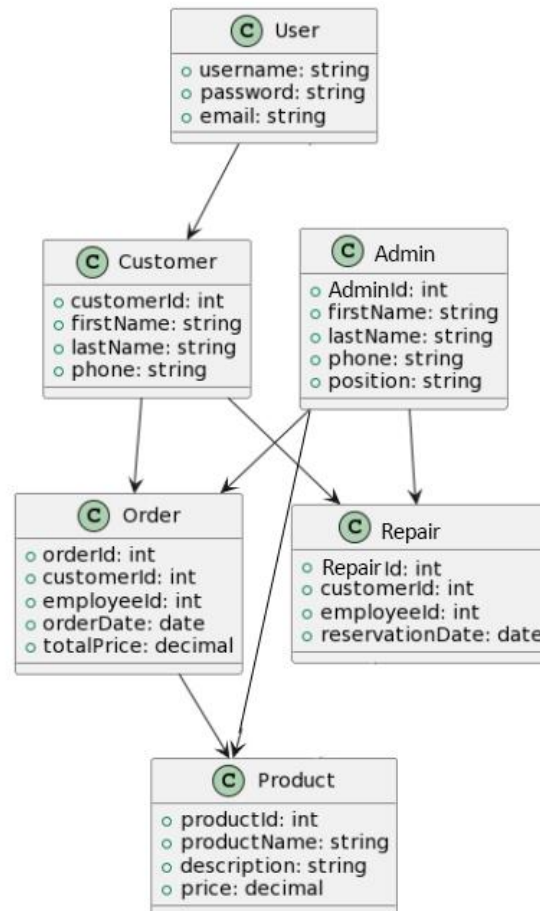


Figure 3.4: Class Diagram

The class diagram includes classes such as "User," "Product," "Order," and "Category," which encapsulate the relevant attributes and methods. The relationships between the classes, such as association, inheritance, and aggregation, are depicted using appropriate connectors. The class diagram highlights the important entities and their interactions within the system, serving as a blueprint for implementing the software architecture. It helps in

understanding the system's design and facilitates efficient code development and maintenance.

3.4.3. Sequence Diagrams

Sequence diagrams provide a visual representation of the interactions between different components and actors in a system. In this project, a set of 15 sequence diagrams have been created to illustrate the flow of actions and messages between the various modules and actors. A sample sequence diagram is shown in figure 3.5.

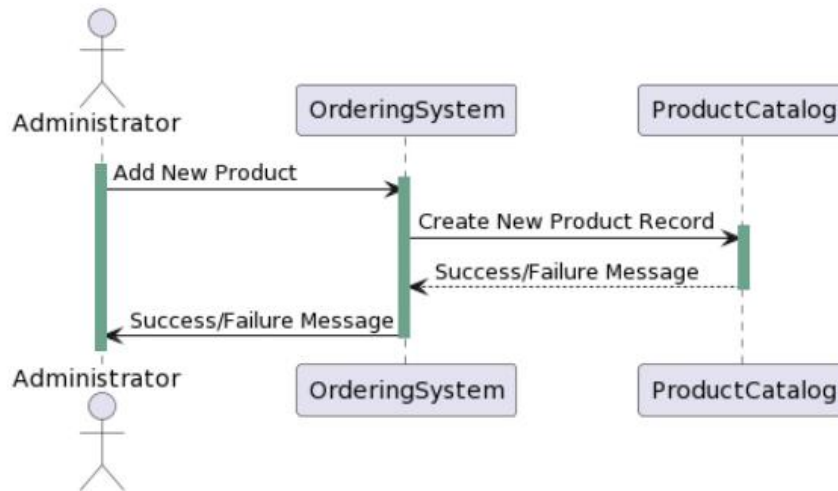


Figure 3.5 : Sequence Diagram for Admin Dashboard

These diagrams depict the dynamic behavior of the system, showcasing how different functionalities and processes are executed. Each sequence diagram focuses on a specific scenario, such as user registration, product ordering, payment processing, and reservation administration, among others. By studying these sequence diagrams, it becomes easier to understand the sequence of events, the order of method invocations, and the data exchanges that occur during the execution of different system features. The sequence diagrams serve as valuable documentation and aid in designing, analyzing, and validating the system's behavior and functionality.

3.5. Data Modelling

Data modeling plays a crucial role in the development of the project by capturing the structure and relationships of the data that will be stored and managed by the system. In this project, a relational data model is employed using the MySQL database management system. The data modeling process involves identifying the entities, attributes, and relationships between them, which are then translated into tables, columns, and foreign key constraints in the database schema. Data modeling ensures data integrity, consistency, and efficiency by defining the appropriate data types, constraints, and normalization techniques. By utilizing a well-designed data model, the project can effectively organize and store data, facilitate data retrieval and manipulation operations, and support the system's functionality in a scalable and maintainable manner.

3.5.1. Entity Relationship Diagram

The entity-relationship (ER) diagram (Figure 3.6) illustrates the logical structure and relationships between the entities in the project's database. It provides a visual representation of how different entities are related to each other and the attributes associated with each entity. The ER diagram showcases the entities involved in the system, such as "Customer," "Product," "Order," and "Category," along with their respective attributes. The relationships between these entities, such as "Customer places Order," "Product belongs to Category," and others, are depicted using appropriate connectors. The ER diagram helps in understanding the database design and provides a blueprint for implementing the data model in the system.

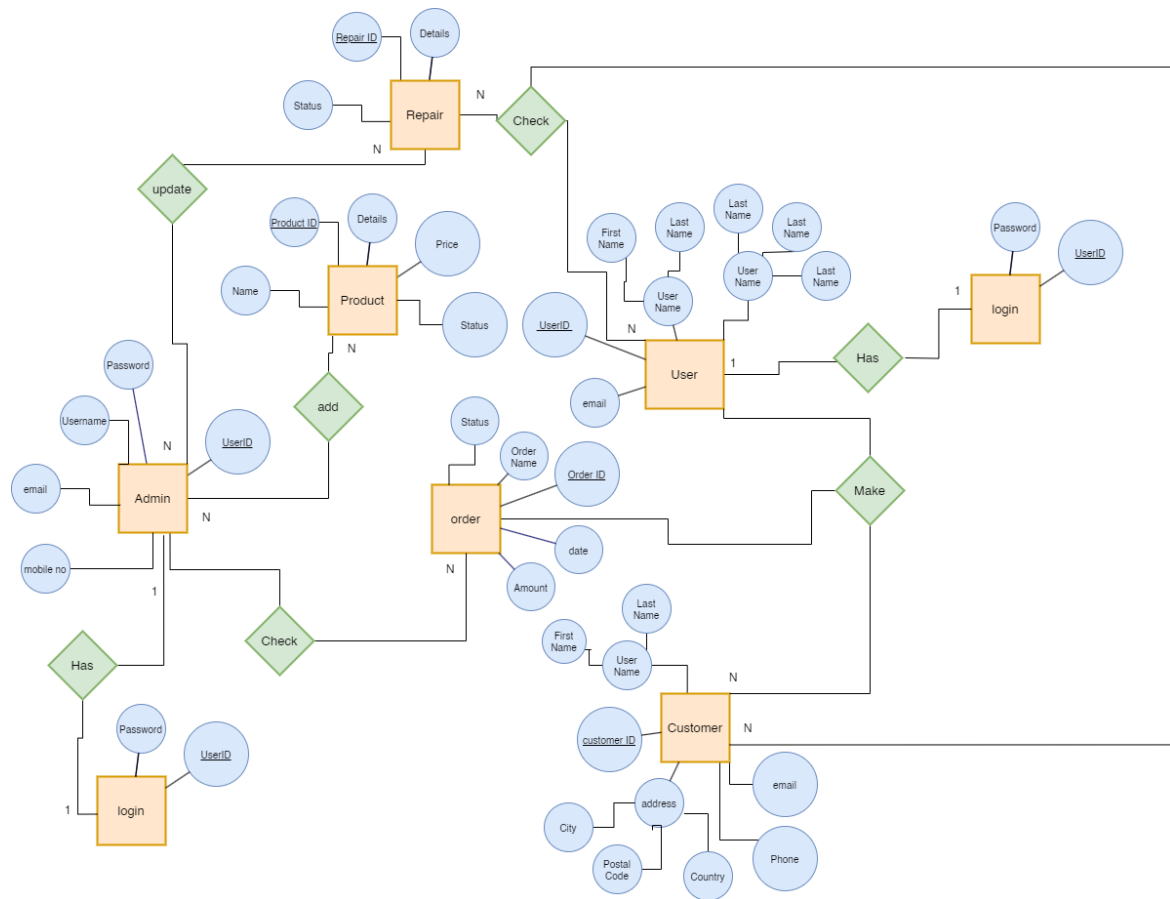


Figure 3.6 : Entity Relationship Diagram

3.6. User Interface Design

In the User Interface Design section, several screenshots of the web pages developed for this project are provided to showcase the visual design and layout of the system. These screenshots (Figure 3.7 to 3.12) offer a glimpse into the user interface and highlight the key components and features of the web application. The screenshots include the homepage, product browsing page, shopping cart, checkout process, PC customization feature, and admin dashboard. Each screenshot captures the aesthetic appeal, user-friendly navigation, and intuitive design elements incorporated in the user interface. These visuals serve as a reference for understanding the look and feel of the system and provide an overview of the user experience. The screenshots demonstrate the efforts made to create an attractive, modern, and responsive user interface that enhances user engagement and facilitates seamless interaction with the system.

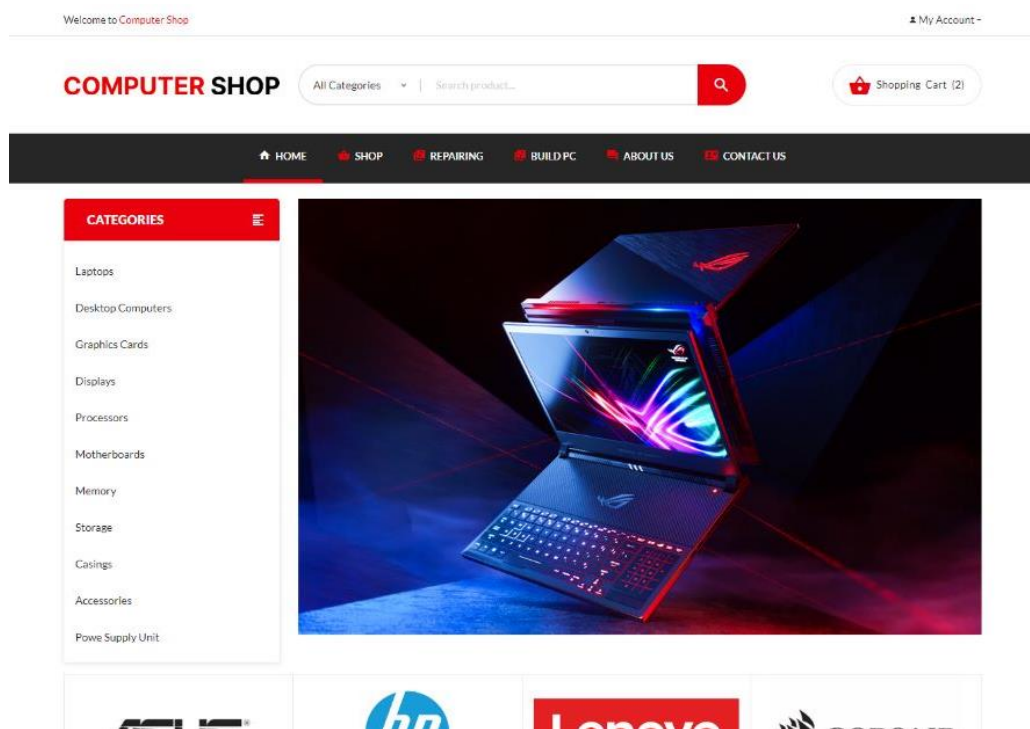


Figure 3.7 : Homepage (Home)

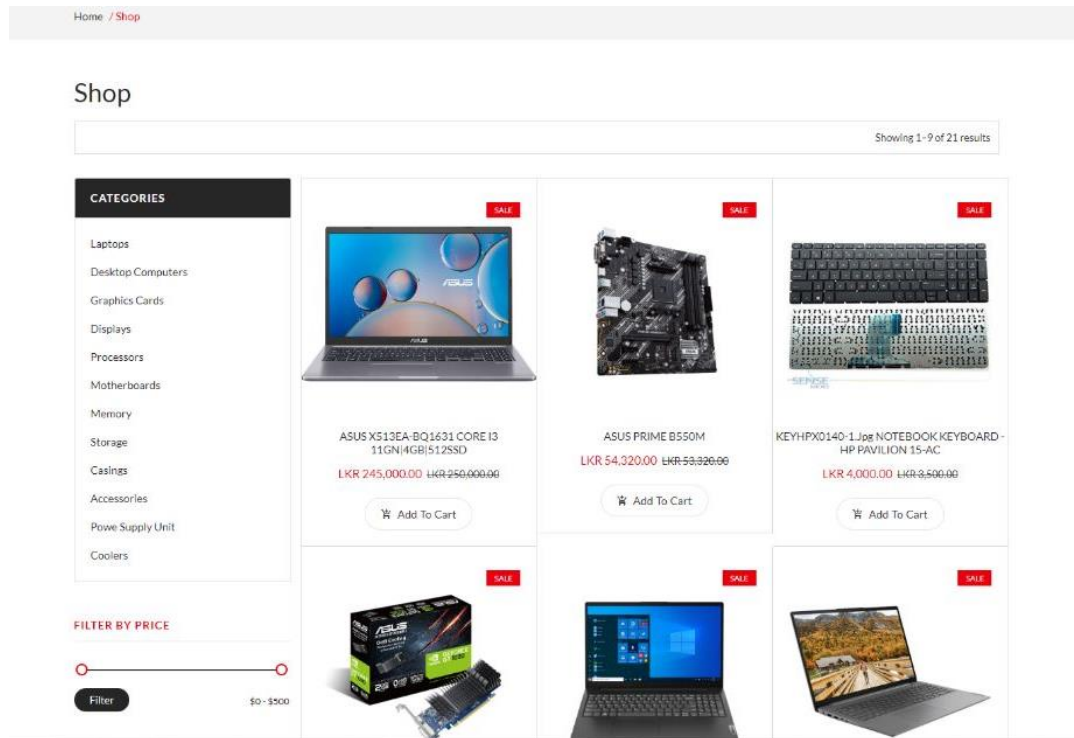


Figure 3.8 : Product Browsing Page (Shop)

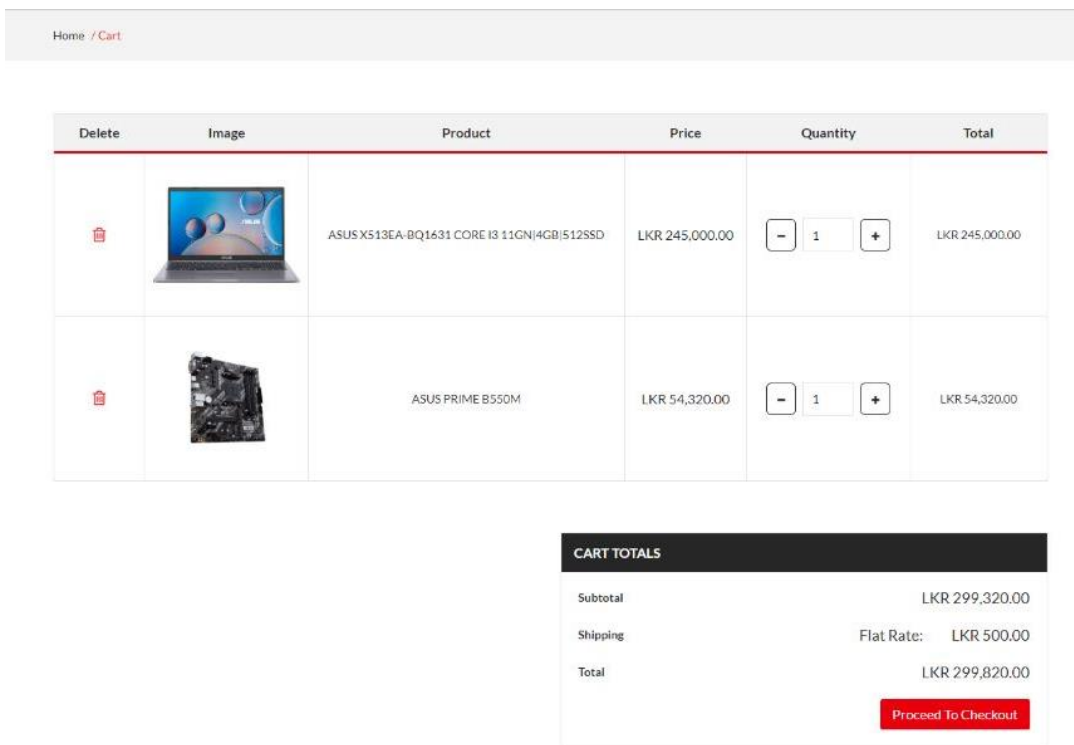


Figure 3.9 : Shopping Cart

[HOME](#)
[SHOP](#)
[REPAIRING](#)
[BUILD PC](#)
[ABOUT US](#)
[CONTACT US](#)

Home / Checkout

BILLING DETAILS

First Name *

Last Name *

Street address *

Town / City *

Postal Code *

State / County *

Phone *

Email Address *

Order Notes

YOUR ORDER

Product	Total
ASUS X513EA-BQ1631 CORE I3 11GN 4GB 512SSD × 1	245000.00
ASUS PRIME B550M × 1	54320.00
Cart Subtotal	LKR 299,320.00
Shipping	LKR 500.00
Order Total	LKR 299,820.00

☐ Card payment
 ☐ Bank transfer

Confirm Order

Figure 3.10 : Checkout Process

Welcome to Computer Shop
 [My Account](#)

COMPUTER SHOP

Search product...

Shopping Cart (0)

[HOME](#)
[SHOP](#)
[REPAIRING](#)
[BUILD PC](#)
[ABOUT US](#)
[CONTACT US](#)

Home / Build Custom PC

Component	Product	Price
CPU	test product	1000
Mother Board	test product	-
Memory 1	- SELECT -	-
Memory 2 (optional)	- SELECT -	-
Storage 1	- SELECT -	-
Storage 2 (optional)	- SELECT -	-

Figure 3.11 : PC Customization (Build PC)

COMPUTER SHOP

Dashboard

Users

Orders

Brands

Categories

Products

View All Products

Add new product

Repairings

Settings

Products

Manage Products.

+ Create

Delete Selected

Show

25 rows

entries

Search:

	IMAGE	NAME	QUANTITY	PRICE	STATUS	ACTION
☐		ASUS X530EA-RQ1631 CORE i3 11G4GB/512SSD	10	250000	ACTIVE	Action
☐		ASUS PRIME 6550M	5	53320	ACTIVE	Action
☐		KEYHPK01401.jpg NOTEBOOK KEYBOARD- HP PAVILION 15-AC	3	3500	ACTIVE	Action
☐		VGA- ASUS GT1030 GEFORCE DDR6 2GB	5	40000	ACTIVE	Action
☐		LENOVO V15 G2TL CORE i5 11G4GB/256SSD/DO5/BLK/MTW- 8AG7y	20	199000	ACTIVE	Action
☐		LENOVO SLIM 3 15ALC6 RYZEN5 5500U/8GB/512SSD/WT1grey/3y	10	235000	ACTIVE	Action
☐		LENOVO SLIM 3 15ALC6 RYZEN7 5700U/8GB/512SSD/WT1grey/3y	1	265000	ACTIVE	Action

Showing 1 to 7 of 7 entries

Previous

1

Next

Figure 3.12 : Admin Dashboard

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Chapter 04 - Implementation

The Implementation section of this thesis presents the details of how the proposed Online Ordering and Reservation System for Computer Store was developed and implemented. This section outlines the step-by-step process followed to transform the design specifications into a fully functional system. It covers the selection and setup of the development environment, the coding and integration of various system components, as well as the testing and deployment of the system. Additionally, this section highlights the technologies and tools utilized during the implementation phase and provides insights into the challenges encountered and the corresponding solutions adopted. By delving into the practical aspects of the project's realization, this section offers a comprehensive overview of the development journey and demonstrates the successful translation of the design into a functioning system.

4.1. Implementation Environment

The implementation environment for the Online Ordering and Reservation System for Computer Store consisted of a set of tools and technologies carefully selected to support the development and deployment of the system. The backend development was carried out using Laravel, a popular PHP framework known for its robustness and scalability. This provided a solid foundation for building the system's server-side components and implementing the business logic. For the front-end development, Vue.js was chosen as the JavaScript framework, offering a reactive and component-based approach to building user interfaces. CSS and HTML were used for styling and structuring the web pages, while jQuery was employed for additional JavaScript functionalities. The MySQL database was utilized to store and manage the system's data, ensuring efficient and reliable data storage. Overall, this combination of technologies formed a powerful and cohesive environment that facilitated the smooth implementation of the Online Ordering and Reservation System.

4.1.1. Hardware Environment

The hardware environment utilized in the development environment was listed in the following configurations.

- Processor: Intel® Core™ i7-8750H
- Storage: 256 GB SSD (OS and software), 1 TB HDD (project files)
- RAM: 32 GB

4.1.2. Software Environment

The software environment utilized in the development environment was listed in the following configurations.

- Microsoft Windows 10: The operating system
- Laravel: PHP framework for backend development.
- Vue.js: JavaScript framework for frontend development.
- HTML: Markup language for creating web pages.
- CSS: Styling language for designing the appearance of web pages.
- JavaScript: Programming language for implementing interactive elements on web pages.
- jQuery: JavaScript library for simplifying DOM manipulation and event handling.
- MySQL: Relational database management system for storing and managing data.
- Apache: Web server software for hosting the application.
- VS Code: Integrated development environment (IDE) for PHP programming.
- Git: Version control system for tracking changes in the source code.

- Composer: Dependency management tool for PHP.
- NPM: Package manager for JavaScript.
- XAMPP: Local development environment for running a web server and database.
- Adobe Photoshop: Image editing software for designing graphical elements.
- Chrome Developer Tools: Browser-based debugging and web development tool.
- Notepad++: Text editor for code editing and manipulation.
- Microsoft Office Suite: Software package for creating documentation and presentations.

These software tools played essential roles in the development and implementation of the Online Ordering and Reservation System, enabling efficient coding, design, database management, testing, and collaboration throughout the project lifecycle.

4.2. Justification For the Implementation Platform

For the implementation of the Online Ordering and Reservation System for Computer Store, a web-based platform was chosen as the most suitable option. This choice takes into account the nature of the system, which involves both client-side and server-side components. In terms of the server-side implementation, two key specifications were considered:

-  Server-side processing framework
-  Database management system

By selecting this implementation platform and the corresponding technologies, the Online Ordering and Reservation System for Computer Store can leverage the benefits of a robust server-side processing framework and a reliable database management system to deliver a seamless and efficient user experience.

4.2.1. Server-side processing framework

In the development of the Online Ordering and Reservation System for Computer Store, careful consideration was given to selecting a suitable server-side processing framework. The server-side, also known as the backend, plays a crucial role in handling database-related operations and responding to end user requests. A server-side framework offers pre-built, tested, and optimized components that can be customized to accelerate the system implementation and minimize code redundancy.

Among the wide range of server-side scripting languages available, PHP emerged as the most fitting choice for this project. Specifically, the Laravel framework was selected due to its extensive features, reliability, and popularity in the industry. Laravel provides a robust foundation for building web applications, offering a comprehensive set of tools and functionalities. It simplifies the development process and enables efficient handling of complex tasks, such as routing, middleware architecture, and validation of front-end requests.

By leveraging Laravel as the server-side processing framework, the Online Ordering and Reservation System for Computer Store benefits from its ease of use, scalability, and compatibility with the PHP language. The familiarity of PHP among developers and the extensive community support further contribute to the project's success. With Laravel as the backbone of the backend, the system can efficiently handle customer requests, process data, and ensure a seamless user experience.

4.2.2. Database management system

In the development of the Online Ordering and Reservation System for Computer Store, a reliable and efficient database management system (DBMS) is crucial for storing and managing the system's data. The selected DBMS for this project is MySQL, a widely used and robust relational database management system developed by Oracle Corporation.

MySQL offers a comprehensive set of features and functionalities that align with the project's requirements. It supports all the necessary CRUD (Create, Read, Update, Delete) operations, allowing end users to easily interact with the database and perform data manipulations. The relational database structure of MySQL enables the organization and categorization of data through tables, schemas, and relations.

One of the notable advantages of MySQL is its scalability and flexibility. It can handle both small and medium-sized enterprises' data requirements efficiently. The ability to create, modify, and maintain data structures seamlessly provides developers with the flexibility needed to adapt to changing business needs. Additionally, MySQL's extensive community support and resources contribute to its reliability and continuous improvement.

MySQL's SQL-based query language provides a familiar and widely adopted standard for interacting with the database. This ensures compatibility with various programming languages and frameworks, including the backend development framework Laravel used in this project. With MySQL as the chosen DBMS, the Online Ordering and Reservation System for Computer Store can store and retrieve data reliably, ensuring the system's smooth operation and efficient data management.

4.2.3. Client-Side Processing Framework

Client-side processing frameworks and libraries play a crucial role in building dynamic and interactive user interfaces for web applications. For the development of our project, we have utilized the following client-side processing frameworks and libraries:

- React

React is a popular JavaScript library used for building user interfaces, particularly for single-page applications. It is maintained by Meta and the community and has a large developer community and extensive online resources for support. React was chosen as the primary client-side processing library due to its maturity, widespread adoption, and the ability to create reusable components that facilitate efficient data updates without requiring a full page reload. It enables us to build a responsive and user-friendly interface for our web application.

- jQuery

jQuery is a lightweight JavaScript library that simplifies DOM manipulation and event handling. It provides a concise and intuitive syntax for performing common tasks, such as traversing and manipulating HTML elements, handling events, and making AJAX requests. Although React is the primary client-side processing library, we have used jQuery to enhance certain functionality and interact with the DOM when needed.

- Vue.js

Vue.js is a progressive JavaScript framework for building user interfaces. It is known for its simplicity, flexibility, and easy integration with existing projects. While React is the main library used in our project, Vue.js has also been utilized for specific components or sections where its features and ecosystem are better suited. This combination allows us to leverage the strengths of both React and Vue.js to create a powerful and versatile user interface.

With React as the primary client-side processing library, supported by jQuery and Vue.js for specific use cases, our project benefits from a robust and efficient approach to handling the view component of the web application. These client-side frameworks and libraries enable us to create a seamless user experience, ensuring smooth interaction and efficient data updates. Combined with Express.js as the backend framework and MySQL as the chosen database management system, we have a comprehensive and well-rounded implementation stack for our project.

4.3. Code & Module Structure Descriptions

The code and module structure of a software project is a critical aspect of its development and maintenance. A well-organized and structured codebase enhances readability, modularity, and scalability, making it easier for developers to collaborate, understand, and maintain the project. In this section, we will delve into the code and module structure of our project, providing an overview of how the codebase is organized, the key modules and components, and the principles and patterns followed to ensure a clean and maintainable codebase. Figure 4.1 shows a diagram describing the interactions between various modules.

By establishing a clear structure and adhering to industry best practices, we aim to create a foundation that promotes code reusability, extensibility, and overall project stability.

4.3.1. User Module

This module handles user-related functionality such as user login and registration. Users can create an account, log in to access their account, and manage their profile information. Sample code is shown in figure 4.1.

```
public function makeLogin(Request $request)
{
    $request->validate([
        'login' => 'required',
        'password' => 'required',
    ]);

    $fieldType = filter_var($request->login, FILTER_VALIDATE_EMAIL) ? 'email' : 'phone';
    request()->merge([$fieldType => $request->login]);

    $credentials = $request->only($fieldType, 'password');
    if (Auth::guard('customers')->attempt($credentials)) {
        return redirect()->intended(route('home'))
            ->withSuccess('Signed in');
    }
    return redirect()->back()
        ->withInput()
        ->withErrors([
            'login' => 'These credentials do not match our records.',
        ]);
}
```

Figure 4.1 : Sample code for User Module

4.3.2. Search Module

The search module allows users to search for specific products or items within the system. It supports search functionality with options for sorting and filtering the search results based on user preferences. Sample code is shown in figure 4.2.

```

public function searchView(Request $request)
{
    $products = Product::with('category', 'brand');
    if($request->has('search')) {
        $products = $products->where('name', 'LIKE', '%'.$request->search.'%');
    }

    $products = $products->where('status', 1)->paginate(20);
    return Inertia::render('SearchResult/Index', ['products'=>$products]);
}

```

Figure 4.2 : Sample code for Search Module

4.3.3. Buy Module

The buy module enables users to purchase products from the system. Users can add items to their cart, proceed to checkout, and complete the purchase. Additionally, the module sends a confirmation message to the user after the checkout process is completed. Sample code is shown in figure 4.3.

```

public function paynow(Request $request)
{
    $app_id = "DQZI118A0338E3A8FA0D2";
    $hash_salt = "XIXC118A0338E3A8FA0F3";
    $app_token = "a9858a20ca41abf0928b1eb45055c56f26a5e92004a95f7c09721c5ab83081b57f7d8efd45309df4.IWLS118A0338E3A8FA106";

    $request->validate([
        'f_name' => ['required'],
        'l_name' => ['required'],
        'email' => ['required', 'email'],
        'phone' => ['required'],
        'address' => ['required'],
        'city' => ['required'],
        'country' => ['required'],
        'zip_code' => ['required'],
        'total' => ['required'],
        'payment_method' => ['required'],
    ], [
        'payment_method.required' => 'Please Select Payment Method'
    ]);

    // dd($request);

    try {
        $cart = Cart::instance('cart')->content();
        $subtotal = (float)(str_replace(", ", "", Cart::instance('cart')->subtotal()));
        $shipping = 500;
        $total = $subtotal + $shipping;

        // dd($cart);

        DB::beginTransaction();
    }
}

```

Figure 4.3: Sample code for Buy Module

4.3.4. Build Your PC Module

This module provides users with the ability to customize and build their own PC. Users can select individual components such as CPU, RAM, storage, etc., and the module helps them create a customized PC configuration based on their preferences. Sample code is shown in figure 4.4.

```
mounted() {
  var self = this;
  $(document).ready(function() {
    console.log($("#cpu-price").text().replace('LKR, '').replace('-', 0'))
  });
  this.getTotal();
  $("#cpu").on("change", function(evt) {
    var id = $(evt.target).val();
    var product = self.products.processors.find(el => el.id = id);
    console.log(product.original_price);
    self.form.cpu.amount = product.original_price
    self.getTotal();
    // if (!Product.selling_price || Product.selling_price == null) {
    //   $('#cpu-price').text(self.price_format(Product.original_price))
    // } else {
    //   $('#cpu-price').text(self.price_format(Product.selling_price))
    // }
  });

  $("#board").on("change", function(evt) {
    var id = $(evt.target).val();
    var Product = self.products.boards.find(el => el.id = id);
    // console.log(Product);
    if (!Product.selling_price || Product.selling_price == null) {
      $('#board-price').text(self.price_format(Product.original_price))
    } else {
      $('#board-price').text(self.price_format(Product.selling_price))
    }
  });

  $("#memory").on("change", function(evt) {
    var id = $(evt.target).val();
    var Product = self.products.memory.find(el => el.id = id);
    // console.log(Product);
```

Figure 4.4 : Sample code for Build PC Module

4.3.5. Repair Module

The repair module allows users to check the status of their product repairs. Users can input their repair request details and track the progress of the repair. Additionally, the module can send an SMS notification to the user upon completion of the repair. Sample code is shown in figure 4.5.

```

<div class="mt-60 mb-5">
  <div class="container p-2">
    <div class="row m-0 p-2">
      <div class="col-lg-2"></div>
      <div class="col-lg-8 p-2"><input type="text" v-model="form.repair_id" placeholder="Enter Your Repairing Number.." class="form-control" /></div>
      <div class="col-lg-2"></div>
      <div class="col-lg-2"></div>
      <div class="col-lg-8"><div class="text-danger">{{ $page.props.errors?.msg }}</div></div>
      <div class="col-lg-2"></div>
      <div class="col-lg-4"></div>
      <div class="col-lg-4 p-2"><Link href="#" @click.prevent="getDetails" class="btn btn-dark w-100">View Repair status</Link></div>
      <div class="col-lg-4"></div>
    </div>
  </div>
  <div class="mt-3">
    <div class="row">
      <div class="col-lg-2"></div>
      <div class="col-lg-8"><div class="m-2 h4 bg-success bg-opacity-25 rounded text-success p-3" v-if="repair[0]?.status == 'completed'">Your rep.
      <div class="col-lg-2"></div>
      <div class="col-lg-2"></div>
      <div class="col-lg-8"><div class="m-2 h4 bg-primary bg-opacity-25 rounded text-primary p-3" v-if="repair[0]?.status == 'pending'">Your repair
      <div class="col-lg-2"></div>
      <div class="col-lg-2"></div>
      <div class="col-lg-8"><div class="m-2 h4 bg-warning bg-opacity-25 rounded text-warning p-3" v-if="repair[0]?.status == 'processing'">Your rep.
      <div class="col-lg-2"></div>
      <div class="col-lg-2"></div>
      <div class="col-lg-8"><div class="m-2 h4 bg-info bg-opacity-25 rounded text-info p-3" v-if="repair[0]?.status == 'closed'">Your repairing co
      <div class="col-lg-2"></div>
      <div class="col-lg-2"></div>
      <div class="col-lg-8"><div class="m-2 h4 bg-danger bg-opacity-25 rounded text-danger p-3" v-if="repair[0]?.status == 'cancelled'">Your repair
      <div class="col-lg-2"></div>
    </div>
  </div>

```

Figure 4.5 : Sample code for Repair Module

4.3.6. Admin Module

The admin module is responsible for managing administrative tasks within the system. It provides functionalities to modify administrative roles, handle product order confirmation, and update the repair status of products. Sample code is shown in figure 4.6.

```

public function update(Request $request)
{
    $request->validate([
        'role_name' => ['required', 'unique:roles,name,' . $request->id],
    ]);

    try {
        DB::beginTransaction();
        $role = Role::find($request->id);
        $role->name = $request->role_name;
        $role->save();
        DB::commit();
        return redirect()->route('settings.roles');
    } catch (Exception $ex) {
        dd($ex);
        DB::rollBack();
        Log::error($ex);
        abort(500);
    }
}

/**
 * Show the form for editing the specified resource.
 *
 * @param int $id
 * @return \Illuminate\Http\Response
 */
public function editPermissions($id)
{
    $role = Role::with('permissions')->find($id);
    $allPermissions = $this->getAllPermissions();
    return Inertia::render('Roles/Permissions', ['role' => $role, 'allPermissions' => $allPermissions]);
}

```

Figure 4.6 : Sample code for Admin Module

4.3.7. Product Module

The product module enables the administration of products in the system. Admin users can add new products, assign them to specific categories and brands, and manage their details such as price, description, and availability. Sample code is shown in figure 4.7.

```

public function store(Request $request)
{
    $request->validate([
        'name' => ['required', 'unique:products,name'],
        'slug' => ['required', 'unique:products,slug'],
        'category' => ['required'],
        'brand' => ['required'],
        'original_price' => ['required'],
        'selling_price' => ['nullable'],
        'qty' => ['required'],
        'description' => ['nullable'],
        'details' => ['nullable'],
        'status' => ['required'],
        'image' => ['nullable', 'max:10000'],
        'other_images' => ['nullable', 'max:10000']
    ]);

    // dd($request->all());

    try {
        DB::beginTransaction();

        $product = new Product();
        $product->name = $request->name;
        $product->slug = $request->slug;
        $product->category_id = $request->category;
        $product->brand_id = $request->brand;
        $product->description = $request->description;
        $product->details = $request->details;
        $product->qty = $request->qty;
        $product->original_price = $request->original_price;
        $product->selling_price = $request->selling_price;
        $product->status = $request->status;
        $product->save();

        DB::commit();
    }
}

```

Figure 4.7 : Sample code for Product Module

4.3.8. Sales Report Module

This module generates sales reports for the system. It provides insights into the overall sales performance, including metrics such as total sales, top-selling products, and revenue analysis. These reports help in monitoring and analyzing the system's sales activities. Sample code is shown in figure 4.8.

```

downloadPDF(id) {
  window.jspdf = window.jspdf.jspdf;
  var doc = new jsPDF('p', 'mm', 'a4');

  doc.html($("#invoice").html(), {
    callback: function(doc) {
      // Save the PDF
      doc.save('invoice-' + id + '.pdf');
    },
    x: 15,
    y: 15,
    width: 180,
    windowWidth: 1024
  });
}

```

Figure 4.8 : Sample code for Sales Report Module

4.4. Acknowledgement Of Reused Codes

In the development process of any project, it is common to leverage existing code resources to enhance efficiency and functionality. This section acknowledges the reuse of codes from external sources that have contributed to the implementation of the proposed system. These reused codes have been appropriately credited to their original authors or organizations. By acknowledging and attributing the reused codes, we recognize the valuable contributions of the developers and the open-source community, fostering a spirit of collaboration and sharing in the software development ecosystem.

Server-Side Frameworks and Libraries

- Laravel: PHP framework for backend development.
- Express.js: Node.js framework for building web applications.
- MongoDB: NoSQL document-oriented database management system.
- MySQL: Relational database management system.
- CodeIgniter: PHP framework for creating web applications.

Client-Side Frameworks and Libraries

- Vue.js: JavaScript framework for building user interfaces.
- React: JavaScript library for building user interfaces.
- jQuery: JavaScript library for simplifying DOM manipulation and event handling.

Chapter 05 - Evaluation

The evaluation section of this project encompasses the comprehensive testing process conducted to ensure the reliability, functionality, and performance of the system. Testing plays a crucial role in validating the implementation and identifying any potential issues or bugs. This section provides a detailed overview of the testing strategies, methodologies, and techniques employed to assess the system's adherence to requirements and its overall quality. The evaluation process involves various types of testing, including unit testing, integration testing, system testing, and user acceptance testing. By conducting rigorous testing, the developer aims to validate the system's functionality, uncover defects, and enhance the user experience.

5.1. Testing

The testing section of this project outlines the comprehensive testing approach undertaken to ensure the quality and reliability of the system. Testing is an essential phase of the development lifecycle as it allows for the identification of any defects, bugs, or inconsistencies in the software. This section provides an overview of the testing methodologies, techniques, and tools used to assess the system's performance, functionality, usability, and security. By conducting thorough testing, the developer aims to validate the system's behavior, ensure its compliance with requirements, and deliver a robust and error-free application.

The testing process for this project followed a systematic and structured approach to ensure the quality and reliability of the system. The testing activities included a combination of manual testing and automated testing techniques.

5.1.1. Unit Testing

In this project, unit testing was an essential part of the development process to ensure the quality and reliability of the codebase. Unit tests were created to evaluate individual units of code, such as functions or methods, in isolation. These tests were designed to verify the

correctness of the code and detect any potential issues or bugs early on. The testing process involved the use of a unit testing framework, such as Jest or PHPUnit, which provided a set of tools and assertions to write and execute the tests. Each unit test focused on a specific functionality or behavior and was written to cover different scenarios and edge cases. By conducting thorough unit testing, developers were able to identify and resolve issues promptly, maintain code integrity, and ensure that each unit of code performed as expected.

5.1.2. Integration Testing

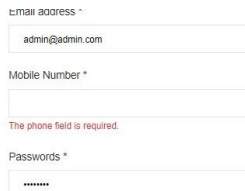
Integration testing in this project was conducted manually to ensure the proper functioning and compatibility of various components and modules when integrated together. The testing process involved systematically combining different modules and verifying their interactions and data flow. Test cases were designed to simulate real-world scenarios and validate the integration points, such as the seamless communication between the user module, search module, buy module, build your PC module, repair module, admin module, product module, and sales report module. During the testing phase, specific inputs and test data were provided to trigger the desired functionalities and validate the expected outcomes. The test cases, as shown in Table 5.1 to 5.4, covered different scenarios and functionalities to validate the integration points. Through this testing approach, the developer systematically examined the interactions between modules and resolved any issues or inconsistencies that arose during the integration process. The manual integration testing helped ensure that the different components of the system worked together seamlessly, providing a cohesive and robust user experience.

Table 5.1 : Test case 01 - Logging to Account (by Customer)

Test Case ID	Test case	Expected Result	Actual Result	Actual Output	Test Status
01	Customer logging with invalid e-mail	An error message should be displayed that “These credentials	An error message was shown that “These credentials do not	 The screenshot shows a login interface with the title 'Login'. It contains two input fields: 'Username or email *' and 'Passwords *'. The 'Username or email *' field contains the text 'imesh'. Below the input fields, a red error message is displayed: 'These credentials do not match our records.'.	Pass

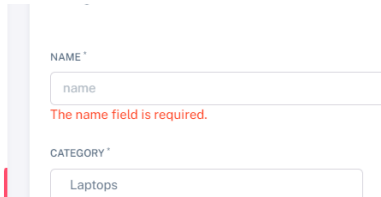
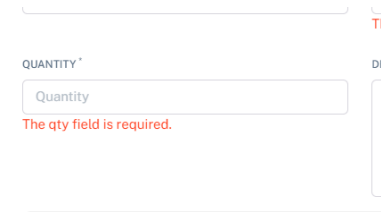
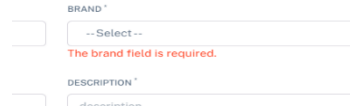
		do not match our records.”	match our records.”		
02	Customer logging with invalid Password	An error message should be displayed that “These credentials do not match our records.”	An error message was shown that “These credentials do not match our records.”		Pass
03	Customer logging with credentials not registered	An error message should be displayed that “These credentials do not match our records.”	An error message was shown that “These credentials do not match our records.”		Pass
04	Customer logging with valid credentials	Customer should redirect to the Home page.	Customer redirected to the Home page.		Pass

Table 5.2 : Test case 02 - Register a account (by customer)

Test Case ID	Test case	Expected Result	Actual Result	Actual Output	Test Status
01	Customer while register with empty field	An error message should be displayed that “The field is empty”	An error message was shown that “The field is empty”		Pass

02	The confirm password field is registered as incorrect	An error message should be displayed that “The password confirmation does not match”	An error message was shown that “The password confirmation does not match”		Pass
03	Customer while register with valid data	Customer should redirect to the home page.	Customer redirected to the home page.		Pass

Table 5.3 : Test case 03 Add New Product (by admin)

Test Case ID	Test case	Expected Result	Actual Result	Actual Output	Test Status
01	User while add new product with “Name” field is empty	An error message should be displayed that “The name field is required”	An error message was shown that “The name field is required”		Pass
02	User while add new product with “Quantity” field is empty	An error message should be displayed that “The qty field is required”	An error message was shown that “The qty field is required”		Pass
03	User while add new product with “Brand” field is empty	An error message should be displayed that “The	An error message was shown that “The		Pass

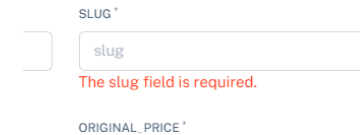
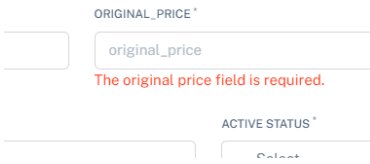
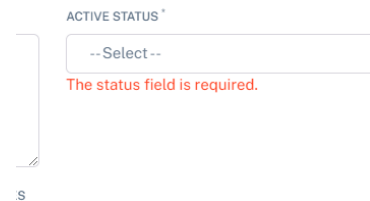
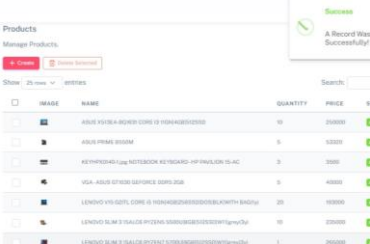
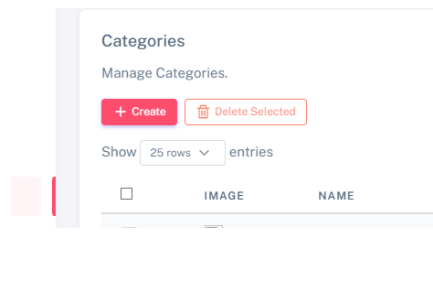
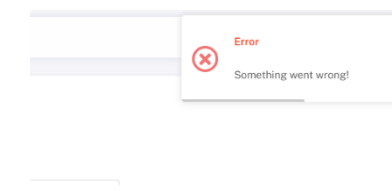
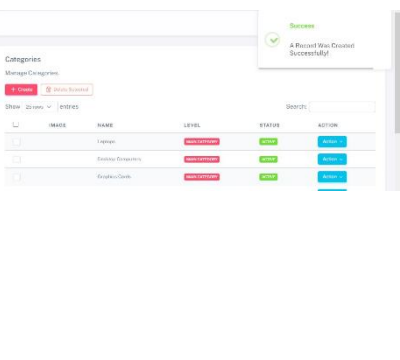
		brand field is required”	brand field is required”		
04	User while add new product with “SLUG” field is empty	An error message should be displayed that “The slug field is required”	An error message was shown that “The slug field is required”		Pass
05	User while add new product with “ORIGINAL_PRICE” field is empty	An error message should be displayed that “The original price field is required”	An error message was shown that “The original price field is required”		Pass
06	User while add new product with “ACTIVE STATUS” field is empty	An error message should be displayed that “The status field is required”	An error message was shown that “The status field is required”		Pass
07	User while add product with valid data and click “Save” Button	A message should be displayed that “A record was create successfull y” and redirect to products page	A message was shown that “A record was create successfull y” and redirect to products page		Pass

Table 5.4 : Test case 04 Add new category (by admin)

Test Case ID	Test case	Expected Result	Actual Result	Actual Output	Test Status
01	Add new category by clicking “create” button	User should redirect to the form of categories	User was redirect to the form of categories		Pass
02	User while add category with empty field	An error message should be displayed that “Something went wrong”	An error message was shown that “Something went wrong”		Pass
03	User while add category with valid data and click “Save” Button	A message should be displayed that “A record was created Successfully” and redirect to categories page	A message was shown that “A record was created Successfully” and redirect to categories page		Pass

5.1.3. System Testing

In this project, system testing was conducted to evaluate the overall functionality and performance of the implemented system. The testing approach involved thoroughly examining the system as a whole, ensuring that all components, modules, and features were working correctly and meeting the defined requirements. Various aspects of the system, such as user interactions, data management, and system behavior, were assessed to ensure their proper functioning. The testing process focused on identifying any potential issues,

such as errors, inconsistencies, or performance bottlenecks, and addressing them promptly. By conducting comprehensive system testing, the developer aimed to validate the system's overall effectiveness and ensure that it delivered the expected outcomes to end-users.

5.1.4. Acceptance Testing

Acceptance testing for the project was conducted by providing users with a user feedback form to gather their input and feedback. Users were given the opportunity to interact with the system and provide their thoughts, comments, and suggestions through the feedback form. Their feedback was carefully analyzed and considered to assess the system's performance and suitability for their needs. This user-centric approach allowed for valuable insights and improvements to be made based on real user experiences, ensuring that the final system met the expectations and requirements of its intended users.

5.2. User Evaluation

The user evaluation phase played a crucial role in assessing the usability and user satisfaction of the implemented system. This section focuses on gathering feedback and insights directly from the end users to evaluate their experiences, identify areas for improvement, and validate the effectiveness of the system's design and functionality. Figure 5.1 shows the part of the user feedback form that was used to collect valuable feedback from users. Through user evaluations and the feedback form, the developer aimed to gain insights into user preferences, challenges, and suggestions for enhancing the system's usability and meeting user needs effectively. By incorporating user feedback into the evaluation process, the developer aimed to ensure that the system aligns with user expectations, offers a seamless and intuitive user experience, and ultimately delivers a valuable solution that meets the needs of its intended users.



User Feedback Form

We would love to hear your thoughts or feedback on how we can improve your experience!

imeshlakshan981@gmail.com [Switch account](#)

Not shared

* Indicates required question

How would you rate your overall experience using the system? *

1 2 3 4 5

☐ ☐ ☐ ☐ ☐

Were you able to easily navigate through the system and find the desired features?

☐ Yes

☐ No

Figure 5.1 : User Feedback Form

The user evaluation results of this project provided valuable insights into the user experience, satisfaction, and areas for improvement. Users expressed positive feedback regarding the system's intuitive interface, ease of navigation, and overall functionality. They found the registration and login process to be straightforward and appreciated the search

module for its efficient filtering and sorting capabilities. The buy module, including the cart and checkout features, received positive feedback for its seamless and secure transaction process. Users also commended the build your PC module for its user-friendly customization options. The repair module was praised for its ability to provide real-time status updates and the convenience of receiving SMS notifications upon completion. In the admin module, users appreciated the flexibility to modify administrative roles and the streamlined product order confirmation process. Figure 5.2 shows the overall user satisfaction.

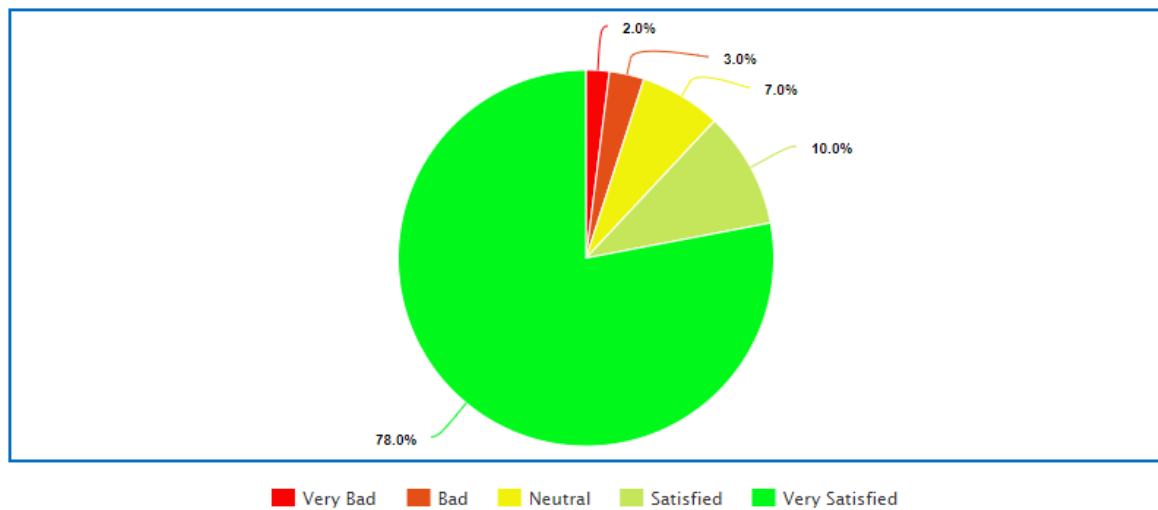


Figure 5.2 : Overall User Satisfaction

Overall, the user evaluation results indicated a high level of user satisfaction, confirming that the system successfully met their expectations and provided a positive user experience. The feedback obtained from users will serve as valuable input for future enhancements and optimizations to further improve the system's usability and meet the evolving needs of its users.

Chapter 06 - Conclusion

The conclusion of the project encompasses a critical evaluation of the entire endeavor, reflecting on the achievements, challenges, and lessons learned throughout the process. The project underwent a rigorous assessment to gauge its effectiveness, usability, and alignment with the defined objectives. Through this evaluation, valuable insights were gained, highlighting both the strengths and areas for improvement. The developer recognizes the importance of continuous learning and growth, and the lessons gleaned from this project will serve as valuable guidance for future endeavors. Moreover, the conclusion also presents opportunities for future enhancements, outlining potential areas where the project can be further developed or expanded to meet evolving user needs and technological advancements. By critically evaluating the project, acknowledging the lessons learned, and identifying avenues for future improvement, the conclusion sets the stage for ongoing progress and success.

6.1.1. Critical Evaluation of the Project

During the critical evaluation of the project, several key aspects were assessed to determine its overall success and effectiveness. One of the primary factors evaluated was the project's ability to meet the defined objectives and deliver the intended outcomes. This involved assessing whether the implemented system fulfilled the functional requirements, provided the desired features and functionalities, and addressed the identified user needs. By scrutinizing these aspects, the developer gained insights into the system's performance and its alignment with the initial project goals.

Another critical aspect evaluated was the usability of the system. This involved examining how intuitive and user-friendly the interface was, and whether users could easily navigate through the system to accomplish their tasks. Usability testing and user feedback played a significant role in this evaluation, as they provided valuable insights into the user experience

and highlighted any areas of difficulty or confusion. The developer used these insights to identify and implement necessary improvements to enhance the system's usability.

Furthermore, the scalability and robustness of the project were evaluated. The system's ability to handle increasing user load, data volume, and concurrent operations was analyzed to ensure its viability in real-world scenarios. Additionally, the project's resilience to errors, exceptions, and security vulnerabilities was scrutinized to identify and rectify any potential weaknesses.

The project's adherence to established coding standards, best practices, and software engineering principles was also critically evaluated. This included assessing the quality of the codebase, the efficiency of the algorithms and data structures used, and the maintainability of the system. By conducting code reviews and employing industry-standard practices, the developer aimed to ensure a high level of code quality and sustainability.

Overall, the critical evaluation of the project provided valuable insights into its strengths, weaknesses, and areas for improvement. These evaluations served as a basis for making informed decisions, addressing shortcomings, and enhancing the project's overall effectiveness and success.

6.1.2. Lessons Learned

The development of this project has provided valuable lessons that have contributed to the growth and improvement of the developer. Throughout the project lifecycle, various skills and knowledge areas were polished, enabling the developer to enhance his professional capabilities. The following are some of the key lessons learned:

Problem-solving and troubleshooting: The project presented numerous challenges that required the developer to think critically and employ problem-solving skills. It helped develop the ability to analyze issues, identify viable solutions, and implement them effectively.

Technical proficiency: The development process provided an opportunity for developer to enhance his technical proficiency in various areas, including programming languages,

frameworks, databases, and software development tools. It allowed them to deepen their knowledge and skills in specific technologies.

User-centric approach: The project emphasized the significance of considering user needs and preferences throughout the development process. It highlighted the importance of gathering user feedback, conducting user testing, and incorporating user-centric design principles to create a system that meets user expectations effectively.

Quality assurance and testing: The project underscored the importance of thorough quality assurance and testing processes. It emphasized the significance of creating comprehensive test plans, conducting rigorous testing, and ensuring the system's reliability, functionality, and performance.

These lessons have not only contributed to the successful completion of the project but also equipped the developer with valuable skills and experiences that can be applied to future projects, promoting continuous growth and improvement.

6.1.3. Suggestions for Future Work

As with any project, there is always room for further improvement and expansion. Based on the experience gained during the development of this project, several suggestions for future work have been identified. These suggestions aim to enhance the system's functionality, user experience, and overall performance. Some of the key suggestions for future work include:

- Implementation of additional features and functionalities to further enhance the system's capabilities.
- Integration of advanced security measures to ensure data privacy and protection.
- Optimization of system performance and scalability to handle larger user loads and accommodate future growth.
- Conducting additional user testing and gathering feedback to further refine the system's usability and address any user concerns.

- Continuous monitoring and improvement of system performance, including regular updates and bug fixes.
- Integration with third-party services or APIs to expand the system's functionality and integration options.
- Exploring opportunities for mobile application development to provide users with a seamless mobile experience.
- Conducting regular maintenance and updates to ensure the system remains up to date with the latest technologies and industry standards.

By considering these suggestions for future work, the system can evolve and adapt to meet the changing needs and expectations of its users, ensuring its long-term success and relevance in the ever-evolving technological landscape.

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Appendix A – System Manual

Hardware requirements

- ✓ A Core i3 8th Gen 1.8GHz or above Processor
- ✓ 8GB or Above RAM
- ✓ 5GB Disk Space (15 GB in SSD is recommended)

Software requirements

- ✓ A coding editor (Visual Studio Code recommended)
- ✓ A modern web browser (Google Chrome or Chromium-based browser recommended)
- ✓ • Any compatible OS (Microsoft Windows 10 or higher version recommended)

Program installation

The following steps must be executed separately for the server, dashboard, and portal to install and run the relevant section for using the proposed system.

1. Copy 'server', 'dashboard', and 'portal' folders to anywhere.
2. Open 'server', 'dashboard' and 'portal' folders separately using a Terminal.
3. Run the 'npm install' command for 'server' and 'yarn install' for 'dashboard' and 'portal'. Wait till dependency installation.
4. Run the 'npm run dev' command for 'server' and 'yarn start' for 'dashboard' and 'portal'.
5. Open a web browser and type 'localhost:3000' in address bar for access the 'dashboard' and 'localhost:3001' for the 'portal'.

Appendix B – User Manual

The User Manual section of this thesis provides a comprehensive guide for users to effectively navigate and utilize the system. It serves as a reference tool, offering step-by-step instructions accompanied by screenshots of various pages and interfaces within the system. By following the user manual, users can familiarize themselves with the system's features, understand its functionalities, and learn how to perform different tasks efficiently. The inclusion of screenshots ensures visual clarity, allowing users to easily identify and follow the instructions provided. Whether users are new to the system or seeking a quick refresher, the user manual serves as a valuable resource for maximizing their experience and proficiency in using the system.

Homepage

The homepage serves as the initial point of entry for users accessing the system. It presents a user-friendly interface with a prominently displayed menu located at the top of the page, as illustrated in Figure B.1. The menu provides intuitive navigation options, allowing users to effortlessly explore different sections of the site. Furthermore, the homepage prominently showcases item prices, enabling users to quickly access pricing information. The inclusion of these features enhances the user experience by providing easy access to relevant content and facilitating efficient navigation throughout the system.

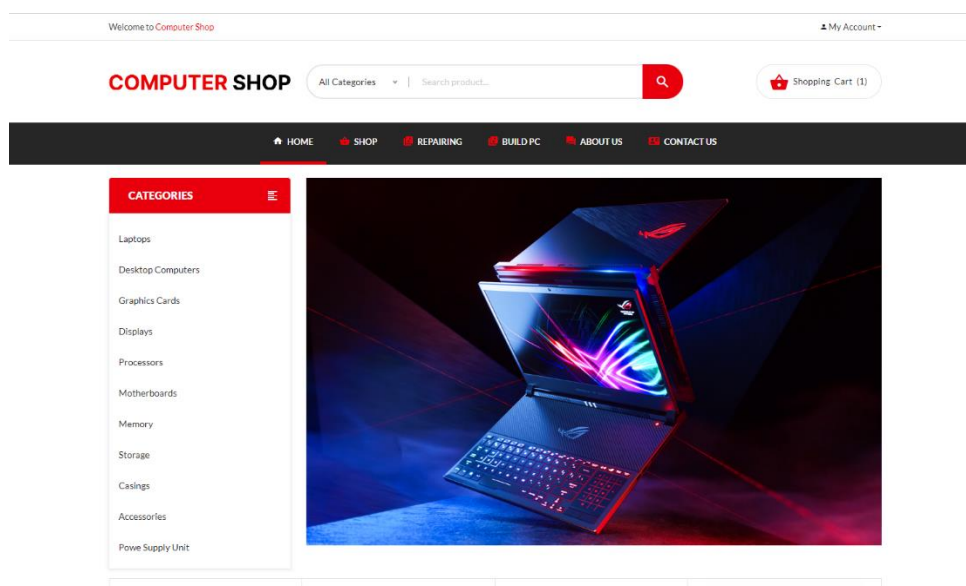
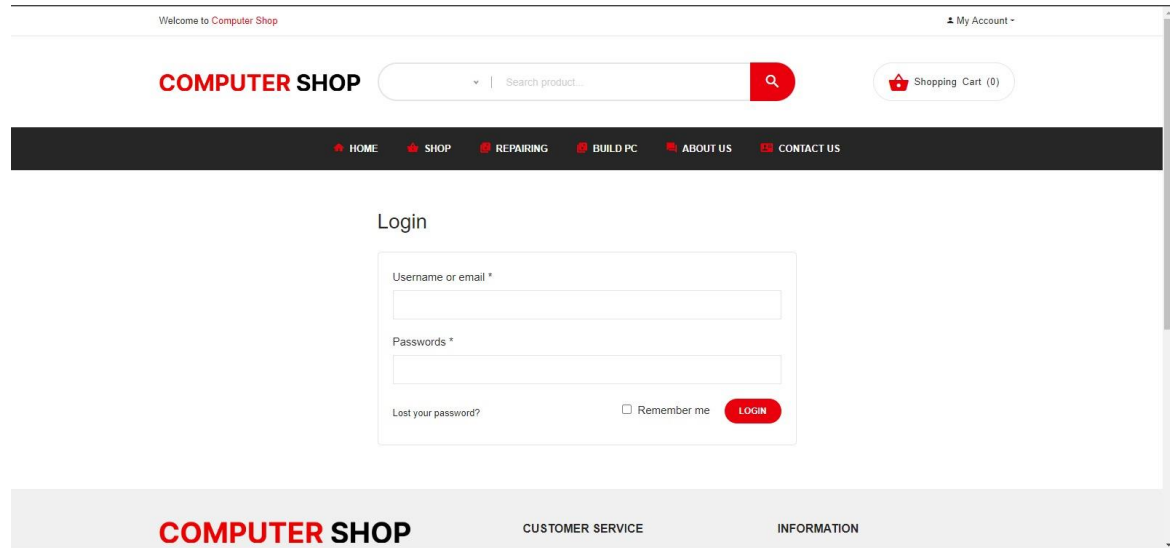


Figure B. 1 : Homepage

Login Screen

The login form, depicted in figure B.2, is an essential component of the login page that grants access to the system's dashboard.



The screenshot shows the login interface of 'COMPUTER SHOP'. At the top, there is a navigation bar with the site name, a search bar, and a shopping cart icon. Below this is a dark navigation menu with links to HOME, SHOP, REPAIRING, BUILD PC, ABOUT US, and CONTACT US. The main content area features a 'Login' form with fields for 'Username or email *' and 'Passwords *'. There is a 'Lost your password?' link and a 'Remember me' checkbox. A red 'LOGIN' button is positioned to the right of the password field. The footer contains the site name, 'CUSTOMER SERVICE', and 'INFORMATION'.

Figure B. 2 : Login Screen

To successfully log in, users must provide valid credentials, including a registered email and password. In the event that any of these fields are left empty or contain incorrect information, access to the dashboard will be denied, and an error message will be displayed. For users who have forgotten their password, there is an option to request a new one by clicking on the "forgot password" link. A password reset email will be automatically generated and sent to the user's registered email address, enabling them to regain access to the dashboard.

Build PC Component

Figure B.3 illustrates the Build Your PC component page, which offers users the flexibility to create a customized computer by selecting various parts and assembling them to build their own PC. This feature empowers users to tailor their computer system according to their specific requirements and preferences. The Build Your PC component page provides an intuitive interface where users can explore a wide range of compatible parts and make informed decisions based on their desired specifications. Additionally, users have the option to obtain a quotation for their custom-built PC before proceeding to the checkout process.

This functionality allows users to review the pricing details and make any necessary adjustments before finalizing their purchase.

Welcome to Computer Shop My Account

COMPUTER SHOP Shopping Cart (0)

[HOME](#) [SHOP](#) [REPAIRING](#) [BUILD PC](#) [ABOUT US](#) [CONTACT US](#)

Home / Build Custom PC

Component	Product	Price
CPU	test product	1000
Mother Board	test product	-
Memory 1	- SELECT -	-
Memory 2 (optional)	- SELECT -	-
Storage 1	- SELECT -	-
Storage 2 (optional)	- SELECT -	-
Casing	- SELECT -	-
Graphics Card	- SELECT -	-
Power Supply Unit	- SELECT -	-
Monitor (Optional)	- SELECT -	-
Mouse (Optional)	- SELECT -	-
Keyboard (Optional)	- SELECT -	-
Sound Devices (Optional)	- SELECT -	-
Total :		00000.00

[Download Quotation](#)
[Add To Cart](#)

COMPUTER SHOP CUSTOMER SERVICE INFORMATION

Figure B. 3 : Build PC Page

Checkout Page

The checkout page, which serves as the final step in the purchasing process. This page provides users with a comprehensive summary of their selected items, including their quantities and corresponding prices. Users can review their order details, shipping address, and payment method before confirming their purchase. The Checkout page ensures a seamless and secure transaction by integrating with trusted payment gateways and encryption protocols. With the user-friendly interface of the Checkout page displayed in Figure B.4, users can conveniently complete their purchase and finalize their order with confidence.

[HOME](#) [SHOP](#) [REPAIRING](#) [BUILD PC](#) [ABOUT US](#) [CONTACT US](#)

Home / Checkout

BILLING DETAILS

First Name *

First Name

Last Name *

Last Name

Street address *

House number and street name

Town / City *

Town / City

Postal Code *

Postal Code

State / County *

State / County

Phone *

Phone

Email Address *

Email

Order Notes

Notes about your order, e.g. special notes for delivery.

YOUR ORDER

Product	Total
ASUS PRIME B550M × 1	54320.00
Cart Subtotal	LKR 54,320.00
Shipping	LKR 500.00
Order Total	LKR 54,820.00

☐ Card payment ☐ Bank transfer

Confirm Order

Figure B. 4 : Checkout Page

Appendix C - Management Reports

The management reports section provides a comprehensive overview of the business through generated reports. These reports offer valuable insights and information that can assist management in making informed decisions and evaluating the performance of the organization. The reports cover various aspects of the business, including sales, revenue, expenses, and other key metrics. By analyzing these reports, management can gain a better understanding of the company's overall performance, identify trends, and make strategic adjustments as necessary. The management reports serve as a valuable tool for monitoring the business's health and facilitating data-driven decision-making processes. Figures C.1, C.2, C.3 show the screenshots of the reports.



Figure C. 1 : Income Report

Orders

Manage Orders.

Show entries Search:

☐	FULL NAME	MOBILE	EMAIL	CITY	TOTAL	STATUS	ACTION
☐	Imesh Imesh	0776093733	imesh1akshan961@gmail.com	Palmadulla	303820.00	PENDING	Action ▾

Showing 1 to 1 of 1 entries Previous **1** Next

Figure C. 2 : Orders Summery

Products

Manage Products.

+ Create Delete Selected

Show entries Search:

☐	IMAGE	NAME	QUANTITY	PRICE	STATUS	ACTION
☐		ASUS X513EA-BQ1631 CORE I3 11GN(4GB/512SSD)	10	250000	ACTIVE	Action ▾
☐		ASUS PRIME B550M	5	53320	ACTIVE	Action ▾
☐		KEYHPX0140-1.jpg NOTEBOOK KEYBOARD -HP PAVILION 15-AC	3	3500	ACTIVE	Action ▾
☐		VGA- ASUS GT1030 GEFORCE DDR5 2GB	5	40000	ACTIVE	Action ▾
☐		LENOVO V15 G2ITL CORE I5 11GN(4GB/256SSD)(DOS)(BLK)(WITH BAG(ty))	20	193000	ACTIVE	Action ▾
☐		LENOVO SLIM 3 15ALC6 RYZEN5 5500U(8GB/512SSD)(W11 grey(3y))	10	235000	ACTIVE	Action ▾
☐		LENOVO SLIM 3 15ALC6 RYZEN7 5700U(8GB/512SSD)(W11 grey(3y))	1	265000	ACTIVE	Action ▾
☐		dsgsd	12	100000	ACTIVE	Action ▾
☐		dsgsdffkqjn	12	100000	ACTIVE	Action ▾

Figure C. 3 : Inventory Report

Glossary

Framework	A framework is a software tool developed to speed up the development process by providing a foundation for solutions, reduce the chance of getting errors, and create a clean, testable code base by providing ready-made, well-tested and customizable components.
Library	A library is a collection of highly used code snippets that might contain prewritten reusable functions, operations, utility programs, packages, and routines.
Operating System	Hardware and end-user communication are allowed, file management, memory management, and process management are performed, input and outputs are handled, and hardware is controlled by an operating system